Does Increasing Government Spending on Tertiary Education Provide a Return on Investment? A Bayesian Causal Inference Analysis

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Abstract

When a national government increases public spending on tertiary (university) education, does that investment provide a positive fiscal return on investment once general economic growth is separated out? Cross-country data make this question difficult to answer because faster-growing economies naturally collect more tax revenue, international indicators divide per-student spending by gross domestic product (GDP) per person and student headcount, and university cohorts take several years to graduate and enter higher tax brackets. We assemble a harmonised five-year panel covering 121 countries from 1985 to 2024 and fit continuous-treatment Multivariate Bayesian Causal Forests that model four interrelated tax changes simultaneously whilst adjusting for pre-treatment country conditions, predicted funding levels, and follow-up macroeconomic growth. Within the central policy range around each income group's benchmark, a +10 percentage-point (pp) increase in per-student university spending as a share of GDP per person shifts the tax mix toward direct income, profit, and capital gains taxation by +0.36 pp after 5 years (with a 93% posterior probability of a positive effect) and +0.71 pp after 10 years (96% probability). Overall tax revenue as a share of GDP takes a full decade to respond as cohorts graduate and enter formal careers, with 10-year total non-resource tax revenue rising by +0.24 pp of GDP (89% probability) and direct income tax revenue rising by +0.12 pp of GDP (90% probability) per +10 pp of per-student spending, by +0.37 pp (100% probability) following five-year within-country funding increases, and by +0.22 pp in total tax (95% probability) and +0.15 pp in direct income tax (98% probability) per +25% increase in real purchasing power parity dollars per student. Translating these 10-year causal slopes into a single-cohort comparison between the multi-year degree cost of educating one entering cohort and that cohort's cumulative tax payments across a 40-year working career reveals a positive expected return on investment across all income groups. In high-income economies, a +10 pp per-student funding boost generates a typical three- to five-fold career return in total non-resource tax revenue (such as a €2.37 billion cohort input in Ireland returning €9.66 billion in total tax with a 77% probability of net profit and €2.87 billion in direct income tax alone, or €11.55 billion in the United Kingdom returning €58.33 billion in total tax) and a four- to twelve-fold return in middle- and low-income economies, whereas expanding the overall university budget without protecting spending per student fails to generate a fiscal return.

Keywords: Multivariate Bayesian Causal Forests; continuous treatment; tertiary education spending; fiscal return on investment; direct income taxation; cross-country panel data.

1 Introduction

Public debates over higher education finance often centre on a simple, practical question. Does government spending on tertiary (university) education provide a positive return on investment to the public purse, or is it a net fiscal cost? At the individual level, decades of labour economics research show that university graduates earn higher wages, experience lower unemployment, and work in larger, more formalised firms than non-graduates (Carneiro et al., 2011; Zimmerman, 2014; Psacharopoulos and Patrinos, 2018;

Deming, 2022). Empirical studies in the United States also show that public investments in college access and per-student instructional resources can recoup a substantial share of their upfront cost over a graduate's working career through higher income tax payments (Trostel, 2010; Chakrabarti et al., 2020; Hendren and Sprung-Keyser, 2020). As university-educated cohorts enter the labour market and raise firm productivity (Moretti, 2004; Valero and Van Reenen, 2019), those private wage gains and corporate spillovers should show up at the national level as stronger personal income, payroll, consumption, and corporate profit tax receipts that pay back the initial university outlay.

Yet working out whether national government spending on tertiary education actually improves a country's tax take is surprisingly tricky. First, if a country's tax revenue rises in the decade after an increase in university funding, we must determine whether universities specifically strengthened the formal tax base or whether the whole economy simply grew richer. If an analyst constructs a national tax-per-person variable by multiplying the tax-to-GDP ratio by GDP per person, any positive effect of education on GDP automatically inflates the tax outcome by mathematical identity. Isolating a genuine fiscal return therefore requires modelling tax shares and tax-to-GDP ratios whilst controlling for starting economic conditions and follow-up economic growth (Imai et al., 2010; Acharya et al., 2016; VanderWeele, 2016). Second, the standard international indicator compiled by the United Nations Educational, Scientific and Cultural Organisation (UNESCO) Institute for Statistics[1] and the World Bank[2] measures public expenditure per university student as a percentage of GDP per person. Because this ratio places both student headcount and GDP per person in the denominator, a low-income country with a tiny elite university sector (such as Malawi or Ethiopia, enrolling under 2% of young adults) can appear to spend over 150% of GDP per person per student, whereas a high-income country enrolling 60% of young adults (such as Denmark or South Korea) appears to spend 27% despite spending nearly three times as many real purchasing power parity (PPP) dollars per student (where PPP adjusts US dollars for differences in local price levels).

A third obstacle is that university investments are continuous policy choices that take time to mature and face diminishing marginal returns. Treating government university spending as a simple binary on/off switch would discard variation in funding intensity and imply that a tiny spending nudge across an arbitrary cutoff produces the same tax return as a major fiscal expansion. At the same time, governments cannot keep increasing university spending indefinitely and expect linear tax gains forever. In our causal framework, we capture non-linearity and diminishing returns in three ways. First, we evaluate local spending changes within a central ±15 percentage-point (pp) policy window around each income group's typical funding level and adjust for each country's predicted spending level $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ given its starting covariates $X_{i,t}$. Second, our regression trees allow the causal slope per +10 pp of funding to vary with enrolment and national income, showing that aggregate tax-to-GDP gains taper off as university systems mature, and third, we verify our results using concave logarithmic funding specifications where each additional dollar of spending yields a smaller marginal return.

Figure 1 summarises our study design for addressing these challenges across 121 countries from 1985 to 2024. On the outcome side (Box 3 of Figure 1), we jointly model four percentage-point tax changes (Y_1, Y_2, Y_3, Y_4) in a single multivariate system at both +5 and +10 years. Specifically, Y_1 is the change in total non-resource tax revenue as a share of GDP (measuring the full Exchequer return across all tax streams, including payroll, consumption, property, and income taxes), Y_2 is the change in direct income, profit, and capital gains tax revenue as a share of GDP (measuring the conservative direct income-tax floor), Y_3 is the change in direct income and profit taxes as a share of total tax revenue, and Y_4 is the change in indirect goods and services (sales and value-added) taxes as a share of total tax revenue. On the treatment side (Box 1 of Figure 1), rather than relying on a single spending definition, we estimate separate models for four continuous university funding variables (Z_1, Z_2, Z_3, Z_4) in turn. Here Z_1 is our primary per-student spending ratio (per +10 pp of GDP per person per student within each World Bank income group), Z_2 is the five-year within-country change in per-student funding (per +10 pp increase), Z_3 is real 2021 PPP dollars per student on a logarithmic scale (per +25% relative increase, removing GDP from the denominator), and Z_4 is the public university budget as a percentage of GDP

[1] UNESCO Institute for Statistics (UIS) education finance database: https://data.uis.unesco.org/.

[2] World Bank World Development Indicators (WDI) portal: https://datatopics.worldbank.org/world-development-indicators/.

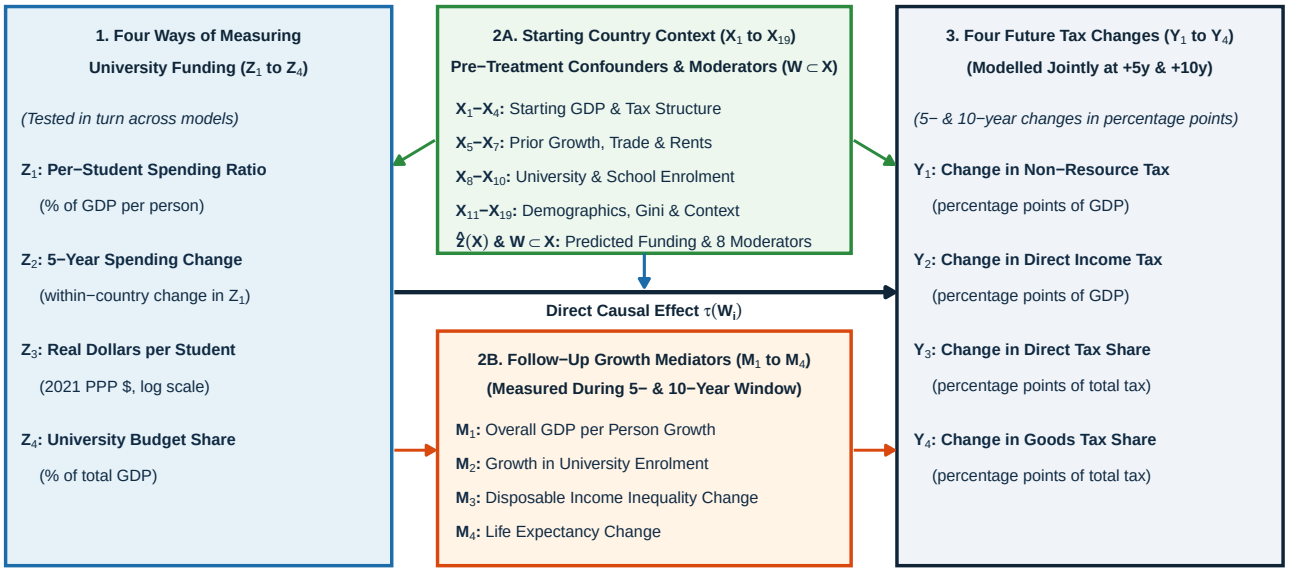


Figure 1: Overview of the study design across 121 countries from 1985 to 2024. Box 1 shows the four continuous university funding measures (Z_1 to Z_4, tested in turn in separate models), Box 2A shows the 19 pre-treatment country covariates (X_1 to X_{19}), predicted funding $\hat{z}(X)$, and the eight treatment-effect moderators ($W \subset X$) that modify the direct causal slope $\tau(W_i)$, Box 2B shows the four follow-up macroeconomic growth mediators (M_1 to M_4), and Box 3 shows the four percentage-point tax revenue changes (Y_1 to Y_4, modelled jointly in every run) over +5 and +10 years.

(per +0.25 pp of GDP, removing student headcount from the denominator). In every model, we control for 19 pre-treatment country characteristics X_1 to X_{19} (Box 2A of Figure 1, of which $W \subset X$ moderate the treatment slope $\tau(W_i)$) and four follow-up macroeconomic growth mediators M_1 to M_4 (Box 2B); throughout the paper, pp denotes percentage points, CI denotes a Bayesian posterior credible interval, and $P = \Pr(\tau > 0 \mid \text{data})$ denotes the posterior probability that the causal slope τ is positive given the data.

We believe this is the first global study to use modern causal inference techniques to test whether public university funding provides a national fiscal return on investment. Existing microeconomic studies estimate fiscal returns by tracking individual college cohorts in a single country such as the United States (Trostel, 2010; Chakrabarti et al., 2020; Hendren and Sprung-Keyser, 2020), which provides detailed lifecycle tax estimates but cannot capture economy-wide corporate spillovers or cross-country differences in tax enforcement. Conversely, sub-national and cross-country macroeconomic studies examine how universities affect GDP per person (Valero and Van Reenen, 2019) or how employment structure affects tax collection (Besley and Persson, 2014; Jensen, 2022) using linear panel regressions that assume constant slopes and analyse one outcome at a time. Unlike single-country lifecycle accounting or linear fixed-effects regressions, our continuous-treatment Multivariate Bayesian Causal Forest approach models all four interrelated tax changes (Y_1 to Y_4) simultaneously across 121 countries while allowing non-linear confounding and heterogeneous returns across development stages. By unbundling the four funding measures (Z_1 to Z_4), separating out follow-up macroeconomic mediators (M), and translating the resulting 10-year causal slopes into single-cohort degree inputs versus 40-year career tax outputs, our approach gives finance and education ministries concrete return-on-investment estimates in their own currency.

The remainder of the paper is organised as follows. Section 2 reviews the literature on tax capacity, university resources, and Bayesian regression tree models for continuous treatments. Section 3 describes the 40-year panel dataset, our four tax outcomes (Y_1 to Y_4), the UNESCO denominator trap across our four funding measures (Z_1 to Z_4), and the country-blocked holdout design. Section 4 explains the continuous-treatment Multivariate Bayesian Causal Forest (MV-BCF) model, why we separate baseline predictors X and growth mediators M from treatment moderators W, and how we account for expected funding levels, with full mathematical derivations in Appendix A. Section 5 presents the empirical results

across time horizons, funding measures, country subgroups, cohort return-on-investment calculations for policy makers, and out-of-sample holdout validation, with additional benchmarks, linear-versus-logarithmic comparisons, and a 20-country policy-maker lookup table in Appendices C to F. Finally, Section 6 discusses multi-cohort fiscal payback timelines, limitations, and directions for future research.

2 Literature Review

In this section we review three strands of research that have largely developed in parallel: the political economy of tax structure modernisation, the economics of per-student university resources, and Bayesian tree ensembles for causal inference. Bringing these three literatures together clarifies why university resources should affect direct income taxation more strongly than consumption taxation. It also explains why flexible nonparametric models are needed to separate continuous funding slopes from general economic growth. Readers interested in comprehensive surveys of each field may refer to Besley and Persson (2014), Deming (2022), and Hill et al. (2020).

2.1 Tax Capacity, Formal Employment, and Tax Structure

A central question in public finance is why high-income countries collect 30% to 45% of GDP in tax revenue, whereas low- and middle-income countries often collect less than 15% of GDP (Besley and Persson, 2009; Gordon and Li, 2009; Besley and Persson, 2014). High-income economies raise most of their revenue through direct personal income taxes, social security contributions, and corporate profit taxes, while developing economies rely heavily on indirect consumption taxes and customs duties. Early cross-country studies emphasised political institutions, property rights, and administrative state capacity (Besley and Persson, 2009; Kaufmann et al., 2011), alongside international accounting standards that separate domestic tax effort from volatile oil and mineral royalties (Prichard, 2016; UNU-WIDER, 2023).[3] Whilst these institutional studies establish that fiscal capacity evolves alongside economic development, they leave open the specific channels through which public investments in skilled workers expand the collectible tax base.

More recent empirical work has focussed on the microeconomic mechanics of tax enforcement, paper trails, and formal employment (Kleven et al., 2016; Keen and Slemrod, 2017; Pomeranz and Vila-Belda, 2019). Kleven et al. (2016) show that modern governments can enforce high direct tax compliance at low audit cost because large, formal firms act as third-party tax collectors. Complex businesses require internal accounting records shared across managers and skilled staff, which makes collusive tax evasion between employers and employees difficult to sustain. Building on this mechanism, Jensen (2022) analyses roughly 100 household surveys across development levels and shows that as the share of salaried employees rises relative to self-employed workers, governments lower income tax exemption thresholds and expand direct income tax collection. Field studies confirm that electronic tax filing and third-party information trails reduce evasion when paired with capable firms and literate taxpayers (Pomeranz and Vila-Belda, 2019; Okunogbe and Pouliquen, 2022), whilst broader direct taxation can also strengthen citizen engagement with the state (Weigel, 2020). By contrast, in developing economies with large informal sectors, indirect consumption taxes such as value-added tax (VAT) remain the main revenue source because formal retail chains only gradually replace informal vendors as incomes rise (Bachas et al., 2024).

This public finance literature suggests a direct connection between university funding and the national tax base that goes beyond headline GDP growth. Universities are themselves large formal employers whose academic and administrative staff pay income tax through automatic payroll withholding. Over the medium term, universities also train the professional, managerial, scientific, and accounting graduates required by formal corporate employers (Kleven et al., 2016; Valero and Van Reenen, 2019). As those graduates enter the workforce, they shift employment into salaried jobs above national income tax exemption thresholds (Jensen, 2022). When this mechanism operates at the national level, higher public

[3] See the UNU-WIDER Government Revenue Dataset (GRD) documentation at https://doi.org/10.35188/UNU-WIDER/GRD-2023.

spending per university student should expand direct income, profit, and capital gains taxation (Y_2 and Y_3) whilst reducing relative reliance on indirect goods and services taxes (Y_4).

2.2 Returns to University Resources and Fiscal Externalities

A large microeconometric literature documents strong private wage returns to university education across both advanced and developing economies (Carneiro et al., 2011; Zimmerman, 2014; Psacharopoulos and Patrinos, 2018; Chetty et al., 2020; Mountjoy, 2022). Reviewing this evidence, Deming (2022) notes that human capital accounts for at least one-third of within-country wage variation and at least half of cross-country income differences. Cognitive and problem-solving skills play a central role in long-run growth (Hanushek and Woessmann, 2012; Angrist et al., 2021), even after accounting for skilled emigration from smaller developing countries (Beine et al., 2008). Beyond individual earnings, university graduates generate productivity spillovers for surrounding workers and firms. Moretti (2004) finds that a higher share of college graduates in a city raises the wages of both graduates and non-graduates, and Valero and Van Reenen (2019) examine roughly 15,000 universities across 1,500 regions in 78 countries and find that university expansions raise regional GDP per person.

Importantly for our study, higher education research demonstrates that graduation rates and subsequent earnings depend heavily on per-student instructional resources rather than enrolment headcounts alone. Bound et al. (2010) and Deming and Walters (2017) show that when public colleges expand student numbers whilst state funding per student stagnates or declines (a phenomenon known as institutional crowding), degree completion rates fall and time to graduation lengthens. Conversely, Chakrabarti et al. (2020) find that increases in state funding for public universities raise bachelor's degree completion and improve students' long-term financial outcomes into their thirties. When graduates complete their degrees and enter formal careers, the fiscal return to the public purse over their working lives can be substantial. Trostel (2010) estimates that lifetime additional income and payroll taxes paid by college graduates exceed direct government subsidies, and Hendren and Sprung-Keyser (2020) show within their Marginal Value of Public Funds framework that direct investments in college access and per-student spending generate positive fiscal externalities through future tax receipts.

2.3 Bayesian Regression Tree Ensembles for Continuous Treatments

Separating the fiscal effect of continuous university funding from general economic development involves two statistical challenges. First, we must control for intermediate economic growth without introducing bias, and second, we must account for non-linear confounding across diverse countries. In cross-country panels, standard linear two-way fixed effects regressions can suffer from negative weighting and functional-form misspecification when treatment slopes vary across countries and over time (Jordà, 2005; Acemoglu et al., 2019; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021). Furthermore, as shown in the causal mediation literature (Imai et al., 2010; Acharya et al., 2016; VanderWeele, 2016; Ting and Linero, 2025), adjusting for follow-up macroeconomic changes (such as GDP per person growth, enrolment growth, inequality, and life expectancy) allows us to isolate the direct structural effect of university spending net of overall economic growth. We also verify this decomposition using two-stage sequential g-estimation (Acharya et al., 2016).

Bayesian Additive Regression Trees (BART; Chipman et al. 2010; Hill 2011; Hill et al. 2020) and Bayesian Causal Forests (BCF; Hahn et al. 2020) provide a flexible nonparametric alternative to linear panel regressions, sitting alongside frequentist causal forests and double machine learning (Chernozhukov et al., 2018; Athey et al., 2019). Tree ensembles automatically capture non-linear relationships and multi-way interactions amongst covariates, with extensions supporting high-dimensional predictors (Linero, 2018), functional smoothing (Starling et al., 2020), count outcomes (Murray, 2021), fast stochastic tree search (Krantsevich et al., 2023), causal mediation (Ting and Linero, 2025), and low-dimensional posterior summaries (Woody et al., 2021). Hahn et al. (2020) show that standard single-forest BART models can suffer from regularisation-induced confounding when strong baseline predictors of the outcome also predict treatment intensity. BCF resolves this problem by splitting the model into a prognostic forest $\mu(\cdot)$ and a separately regularised treatment-effect forest $\tau(\cdot) \cdot Z_{i,t}$, with recent extensions to multivariate

outcomes and longitudinal panels (McJames et al., 2024, 2025). Because $Z_{i,t}$ enters our structural equation in Section 4 (and Appendix A) multiplicatively as a scalar multiplier on the four-dimensional leaf vector $\tau_{l,b} \in \mathbb{R}^4$, the MV-BCF algebra extends directly to a continuous treatment variable $Z_{i,t} \in \mathbb{R}$ by replacing the binary propensity score with the expected continuous funding level $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ inside $\mu(\cdot)$, mirroring continuous-treatment residualisation in causal forests and double machine learning (Robinson, 1988; Hirano and Imbens, 2004; Chernozhukov et al., 2018; Athey et al., 2019).

Readers will note that the Multivariate Bayesian Causal Forest (MV-BCF) method used in this paper (McJames et al., 2025) was co-developed by the author with Nathan McJames, Ann O'Shea, and Yong Chen Goh. Given this connection, we want to make it clear from the outset that this paper is not written to promote one causal estimator over all others. In Section 5.5, we benchmark the out-of-sample predictive accuracy of continuous MV-BCF against single-forest BART, random forests, historical training means, ridge regression, zero-change persistence, and linear local projections. In Appendix C, we also compare its causal slope estimates directly against separate univariate BCFs, single-forest BART, linear local projections, two-way fixed effects, cross-fitted double machine learning, and frequentist orthogonal causal forests (Athey et al., 2019). Throughout the paper, we discuss the limitations of the BCF approach alongside its strengths.

3 Data

3.1 Global Five-Year Panel Structure and Country-Blocked Holdout Split

Annual macroeconomic series for education spending and tax revenue contain frequent reporting gaps and short-term business-cycle fluctuations. Following standard practice in empirical macro-panel studies (Acemoglu et al., 2019), we average annual observations from six international databases into eight non-overlapping five-year waves spanning 40 years: `1985-1989`, `1990-1994`, `1995-1999`, `2000-2004`, `2005-2009`, `2010-2014`, `2015-2019`, and `2020-2024`. These sources are the World Bank WDI, the UNESCO Institute for Statistics, the International Monetary Fund (IMF) Fiscal Monitor,[4] the Worldwide Governance Indicators (WGI), the Standardised World Income Inequality Database (SWIID version 9.6; Solt 2020),[5] and the Organisation for Economic Co-operation and Development (OECD) *Education at a Glance* financing taxonomy.[6] Constructing five-year waves matches the typical duration of an undergraduate degree programme and reduces serial correlation from annual budgetary carryover.

Although the full World Bank register lists 217 sovereign territories, many small island states and conflict-affected territories do not report tertiary education expenditure or disaggregated tax revenues to UNESCO and the World Bank. In addition, raw international fiscal and education archives occasionally contain reporting anomalies where currency-unit mismatches, decimal-place errors, or offshore sovereign resource funds distort headline ratios. Screening out impossible tax-to-GDP values above 60% or below 0.5% of GDP and near-zero tertiary reporting glitches below 0.5% of GDP per person removes five raw reporting anomalies in the pipeline: Sudan in `1995-1999` (recorded at 629.2% of GDP in tax due to a currency-unit mismatch), Timor-Leste in `2010-2014` (recorded at 110.0% of non-oil GDP due to offshore petroleum fund accounting), the United Arab Emirates in `2010-2014` and `2015-2019` (recorded below 0.4% of GDP in central federal tax), and Madagascar in `2015-2019` (recorded at 0.036% of GDP per person per student due to a decimal-place glitch). Across the full cleaned panel before requiring forward tax changes, 808 five-year country-waves have observed university spending or tax structure data (summarised descriptively in Table 1). Requiring strictly observed per-student university funding (Z_1 or Z_3) and all four forward tax changes (with zero imputation on treatments or outcomes) yields 371 country-waves across 121 countries at the +5-year horizon ($t \to t+1$, waves 1 to 7) and 283 country-waves across 112 countries at the +10-year horizon ($t \to t+2$, waves 1 to 6), while requiring two consecutive waves of per-student funding for Z_2 yields 176 waves across 86 countries at +10 years and requiring aggregate tertiary budget data without student headcounts for Z_4 yields 403 waves across 123 countries

[4] IMF Fiscal Monitor database: `https://www.imf.org/en/Publications/FM`.
[5] SWIID version 9.6 database and documentation: `https://fsolt.org/swiid/`.
[6] OECD *Education at a Glance* portal: `https://www.oecd.org/en/publications/education-at-a-glance.html`.

at +10 years (with Z_4 also verified on the overlapping $N = 272$ sample in Check R17 of Appendix Table 5).

To evaluate how well our continuous-treatment models generalise to unseen countries, we partition the countries using a deterministic 5-fold country-blocked split stratified by each country's modal World Bank income group (sorting countries alphabetically by ISO-3 code within each modal income group and assigning folds $1, \ldots, 5$ sequentially). Every wave of a given country is assigned to the same fold so that no country ever straddles the training and test sets. Folds 1 to 4 (80% of countries: 98 countries and 301 country-waves at +5 years; 90 countries and 229 country-waves at +10 years) form the training sample used to compute income-group funding medians and covariate centring statistics without test-set leakage, whereas Fold 5 (20% of countries: 23 unseen countries and 70 country-waves at +5 years; 22 unseen countries and 54 country-waves at +10 years) serves as a locked out-of-sample holdout test set evaluated in Section 5.5. Occasional missing values in auxiliary covariates inside $X_{i,t}$ and $W_{i,t}$ (such as baseline gross tertiary enrolment in 39 of the 283 ten-year waves where UNESCO records per-student expenditure ratios without publishing the separate enrolment headcount) are imputed using strictly out-of-fold training medians so tree splits can be evaluated on all 283 waves, and we confirm in Check R4 of Appendix Table 5 that dropping all incomplete covariate rows ($N = 238$) leaves our results unchanged. In Sections 5.1 to 5.3, we report posterior treatment slopes from the full 5-fold country-blocked cross-fitted panel ($N = 371$ at +5 years and $N = 283$ at +10 years, where $\hat{z}(X_{i,t})$ is predicted strictly out-of-fold across the five country folds), whilst in Section 5.5 and Check R13 we fit both the random forest for $\hat{z}(X_{i,t})$ (using 4-fold internal cross-fitting across Folds 1 to 4) and the MV-BCF model exclusively on the 80% training countries to generate strictly out-of-sample predictions on the 22 locked holdout countries in Fold 5.

3.2 Tax Revenue Outcomes (Y_1 to Y_4)

To examine how university funding affects both overall tax collection and the mix of taxes without introducing mechanical GDP product identities, we distinguish baseline tax levels $L_{1,i,t}, \ldots, L_{4,i,t}$ at wave t from our four percentage-point forward change outcomes $Y_{k,i,t+h} = L_{k,i,t+h} - L_{k,i,t}$ ($k \in \{1, 2, 3, 4\}$) over +5 years ($h = 1$) and +10 years ($h = 2$). Our first outcome, $Y_{1,i,t+h} = L_{1,i,t+h} - L_{1,i,t}$, is the change in total non-resource tax revenue as a percentage of GDP (ΔTax/GDP, in pp of GDP), based on the World Bank WDI tax-to-GDP series (`GC.TAX.TOTL.GD.ZS`). Following the fiscal-capacity literature (Prichard, 2016; UNU-WIDER, 2023), in major oil- and mineral-exporting country-years where natural resource rents (`NY.GDP.TOTL.RT.ZS`) exceed 10% of GDP, we subtract half of the excess resource rents at the annual level before five-year wave averaging ($L_{1,i,\mathrm{yr}} = \mathrm{Tax/GDP}_{i,\mathrm{yr}} - 0.5 \max(0, \mathrm{Rents/GDP}_{i,\mathrm{yr}} - 10)$) as a harmonised proxy so that volatile commodity royalties do not masquerade as domestic tax effort. Whilst the UNU-WIDER Government Revenue Dataset provides consolidated general-government non-resource tax series for a subset of waves, we verify in Appendix C (Check R1) that using raw, unadjusted WDI tax-to-GDP yields nearly identical slopes.

Our second outcome, $Y_{2,i,t+h} = L_{2,i,t+h} - L_{2,i,t}$, is the change in direct income, profit, and capital gains tax revenue as a percentage of GDP (ΔIncomeTax/GDP, in pp of GDP), which captures revenue collected directly from individual wages, salaries, and corporate profits. Our third outcome, $Y_{3,i,t+h} = L_{3,i,t+h} - L_{3,i,t}$, is the change in direct income, profit, and capital gains taxes as a percentage of total tax revenue (`GC.TAX.YPKG.ZS`, ΔIncomeTax/Rev, in pp of revenue), which measures structural modernisation toward progressive direct taxation (Besley and Persson, 2014; Jensen, 2022). Our fourth outcome, $Y_{4,i,t+h} = L_{4,i,t+h} - L_{4,i,t}$, is the change in indirect goods and services taxes (such as VAT, sales taxes, and excise duties, `GC.TAX.GSRV.RV.ZS`) as a percentage of total tax revenue (ΔGoodsTax/Rev, in pp of revenue; Bachas et al. 2024). Readers should note that in levels at any wave t, direct income tax as a share of GDP satisfies $L_{2,i,t} = (L_{1,i,t} \times L_{3,i,t})/100$, so for a marginal funding change the first-difference slopes satisfy $\tau_{Y_2} \approx (\bar{L}_1 \tau_{Y_3} + \bar{L}_3 \tau_{Y_1})/100$. Thus, Y_1, Y_3, and Y_4 capture the two underlying policy dimensions (whether overall tax collection rises and whether the tax mix shifts from consumption taxes toward income taxes), whilst Y_2 measures their combined impact on the national budget, and our Multivariate Bayesian Causal Forest models all four change outcomes (Y_1 to Y_4) simultaneously through a joint 4×4 covariance matrix Σ.

3.3 The UNESCO Denominator Trap and Four University Funding Measures (Z_1 to Z_4)

The standard international indicator of per-student university spending compiled by UNESCO and the World Bank divides public spending per university student by GDP per person. As shown in Equation (1), this ratio is algebraically identical to dividing the public university budget (% of GDP) by the number of university students per person:

$$\text{UNESCO Ratio}_{i,t} = \frac{\text{University Spend per Student}_{i,t}}{\text{GDP per Person}_{i,t}} \times 100 = \frac{\text{University Budget (\% GDP)}_{i,t}}{\text{Students per Person}_{i,t}}. \quad (1)$$

Because both GDP per person and student headcount appear in the denominator, countries with very different university systems can record similar headline ratios for completely different reasons. To see the raw data before any statistical modelling, Figure 2 plots the observed distributions of the UNESCO per-student ratio (used to construct Z_1), real 2021 PPP dollars per student (used to construct Z_3), gross tertiary enrolment, and the direct income tax share (Y_3) across the four World Bank income groups. Table 1 reports the corresponding group summaries.

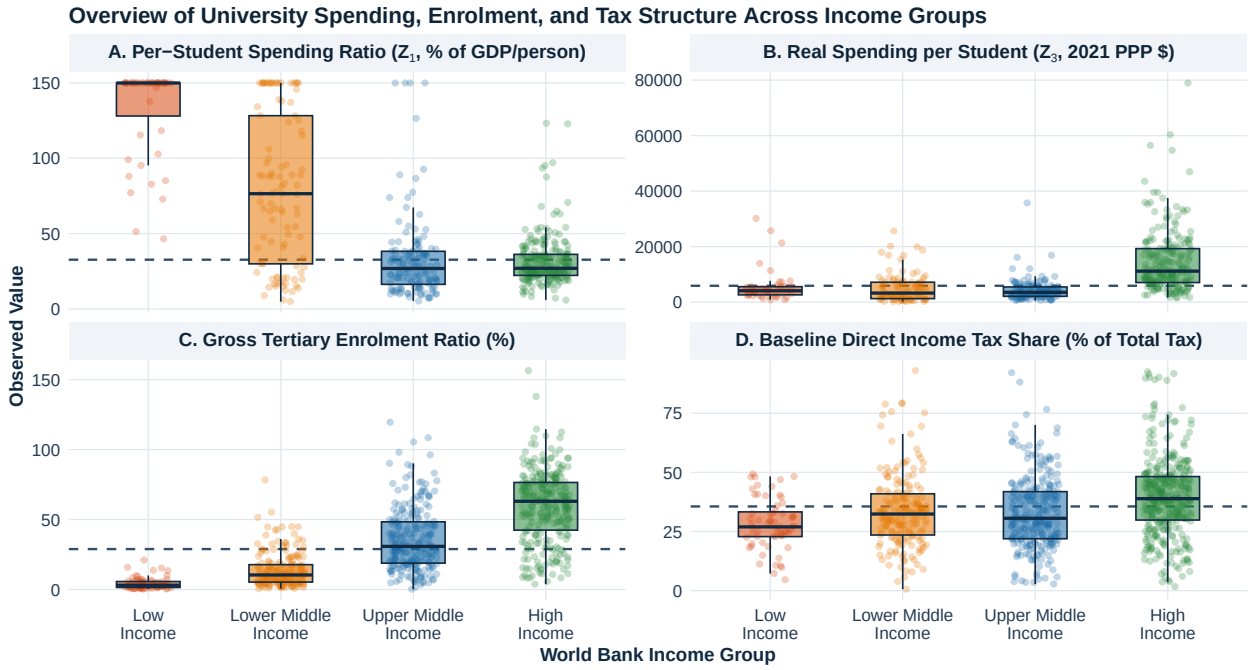


Figure 2: Observed distributions of (A) the UNESCO per-student spending ratio (% of GDP per person, used for Z_1 and winsorised at 150% for display), (B) real spending per university student in 2021 PPP dollars (used for Z_3), (C) gross tertiary enrolment ratio (%), and (D) baseline direct income and profit tax share (% of revenue, Y_3) across the four World Bank income groups. Horizontal dashed lines mark the global median (or mean in Panel D).

Table 1: University spending indicators, enrolment, and baseline direct income tax share (L_3) across World Bank income groups in the full cleaned descriptive panel ($N = 808$ five-year waves across 185 country-by-income-group appearances prior to requiring forward tax changes and student headcounts; individual countries that move across World Bank income thresholds over the 40-year window appear in more than one row). Group medians are reported for skewed spending and enrolment variables and group means for the baseline direct income tax share.

Income Group (Countries, Waves)	UNESCO Ratio (%)	Real PPP $ / Student	Univ. Budget (% GDP)	Tertiary Enrol. (%)	Direct Income Tax Share (%)
Low income (21, 80)	**205%**	$3,957	0.47%	2.3%	29.2%
Lower middle (41, 172)	76.5%	$3,258	0.65%	7.4%	30.8%
Upper middle (54, 219)	26.8%	$3,521	0.64%	25.5%	33.1%
High income (69, 337)	26.9%	**$11,148**	**0.94%**	**58.3%**	**38.4%**

Figure 2 and Table 1 show clearly how the UNESCO denominator trap works in practice. In low-income countries (such as Uganda, Malawi, or Ethiopia), only 2.3% of young adults attend university. Dividing a small university budget (0.47% of GDP) by a tiny student headcount and a low GDP per person produces a raw median UNESCO ratio of 205%, which we cap at 150% in Figure 2A to make the plot easier to read. That tiny elite student body also explains why median real PPP spending per student in low-income countries ($3,957, concentrated in one or two national universities) is slightly higher than in lower-middle-income countries ($3,258), where enrolment is over three times higher (7.4% median) and budgets are stretched across many more students. Once we unbundle the denominators in Table 1, high-income countries (such as Denmark, Norway, or South Korea) spend nearly three times as much per student in real purchasing power ($11,148 vs $3,957) and twice as much of their national income (0.94% vs 0.47% of GDP), whilst collecting a larger share of their tax revenue from direct income and profit taxes (38.4% vs 29.2%).

Because comparing raw UNESCO ratios across income groups would conflate small elite enrolment with high per-student funding, we construct our primary continuous funding variable Z_1 (which we refer to as Scaling A) within each World Bank income group. Specifically, we centre each country-wave's winsorised per-student spending ratio on the training-set median of its own income group ($m_{g(i,t)} = 150.0\%$ for low income after the 150% pre-winsorisation cap vs 183.8% raw, 68.9% for lower-middle income, 27.6% for upper-middle income, and 26.4% for high income) and divide by 10 percentage points. To ensure our causal slope reflects realistic policy changes rather than linear extrapolation to extreme ratios, we winsorise Z_1 at ± 1.5 units (± 15 pp around the group median) so that $Z_{1,i,t} \in [-1.5, +1.5]$ is centred at zero, where $Z_{1,i,t} = 0$ represents the income-group median benchmark and a unit change ($\Delta Z_1 = +1.0$ or -1.0, $\mathrm{SD}(Z_1) = 1.05$) represents a $+10$ pp increase or -10 pp decrease in per-student funding (we also compare Z_1 against a logarithmic $+25\%$ transformation of the same ratio, termed Scaling B, in Appendix E). Readers should note that because more than half of low-income country-waves exceed the 150% pre-winsorisation cap ($m_{\mathrm{Low}} = 150.0\%$), $Z_{1,i,t}$ in the low-income stratum varies over $[-1.5, 0]$ below the cap, so we verify in Check `R1b` (omitting the 150% cap), Scaling B (using raw $m_{\mathrm{Low}} = 183.8\%$), and Z_3 (real PPP dollars) that preserving upper-tail variation yields virtually identical conclusions.

On this zero-centred Z_1 scale, Nordic and continental European high-income countries with generous public university funding sit near the top of the distribution (Norway, Sweden, and Denmark at $+1.2$ to $+1.5$, and Austria at $+0.9$ to $+1.0$). Countries near their group median sit near 0 (New Zealand and Iceland at -0.03), whereas countries relying more heavily on private tuition sit below 0 (Spain at -0.40, the United States at -0.59, Chile at -0.69, and South Korea at -1.15). Ireland illustrates a striking chronological trajectory along Z_1: it sat above the high-income median in `1995-1999` (28.2%, $Z_1 = +0.18$), dipped slightly below in `2000-2004` (25.6%, $Z_1 = -0.08$), recovered above the median in `2005-2009` (28.4%, $Z_1 = +0.20$) and `2010-2014` (27.6%, $Z_1 = +0.12$), and then dropped sharply to 17.3% ($Z_1 = -0.91$, or 9.1 pp below the median) in `2015-2019` as student numbers grew much faster than state grants after the 2008 financial crisis. Alongside Z_1, we construct three additional zero-centred continuous funding variables tested in turn in Section 5.2 (each winsorised at ± 1.5 units so $Z \in [-1.5, +1.5]$): $Z_2 = \mathrm{clip}(\Delta\mathrm{Spend}_{i,t}/10, -1.5, +1.5)$ measures within-country five-year per-student funding changes centred at zero change (per $+10$ pp expansion), $Z_3 = \mathrm{clip}\big(\log(\mathrm{PPP\$}_{i,t}/m^{\mathrm{PPP}}_{g(i,t)})/\log(1.25), -1.5, +1.5\big)$ measures log real 2021 PPP dollars per student centred on the training income-group median (per $+25\%$ relative dollar increase), and $Z_4 = \mathrm{clip}((\mathrm{BudgetGDP}_{i,t} - m^{\mathrm{Bud}}_{g(i,t)})/0.25, -1.5, +1.5)$ measures the total public university budget as a share of GDP centred on the training income-group median (per $+0.25$ pp of GDP).

4 Methodology

In an ideal experiment, we would randomly assign different countries to increase per-student university funding by varying amounts and then track their tax receipts over the next decade. In observational data, however, governments set their university budgets in response to economic conditions, demographics, and existing tax capacity. The relationships among these macroeconomic variables are also strongly non-linear (Hill, 2011; Hill et al., 2020). To separate the causal slope of continuous university funding

from confounding macroeconomic trends whilst modelling all four interrelated tax outcomes (Y_1 to Y_4) simultaneously, we use the continuous-treatment Multivariate Bayesian Causal Forest (MV-BCF) model (Hahn et al., 2020; McJames et al., 2025). We present the intuitive setup of the model here and provide full mathematical derivations in Appendix A.

The Multivariate Bayesian Causal Forest (Hahn et al., 2020; McJames et al., 2025) splits the prediction of future tax changes into two distinct regression tree ensembles (Chipman et al., 2010). The first ensemble estimates a baseline prognostic trajectory $\mu(\cdot)$ (what would have happened to a country's tax take naturally when $Z_{i,t} = 0$, given its 19 pre-treatment country characteristics $X_{i,t}$, its predicted funding level $\hat{z}(X_{i,t})$, and its four follow-up macroeconomic growth mediators $M_{i,t\to t+h}$), whilst the second ensemble estimates a causal treatment slope $\tau(W_{i,t})$ that measures the effect of a unit change in university funding $Z_{i,t}$ and can vary with a vector of eight pre-treatment structural moderators $W_{i,t} = (W_{1,i,t}, \ldots, W_{8,i,t})^\top \subset X_{i,t}$ (a complete data dictionary defining every variable in Y, Z, X, W, and M is provided in Appendix B, Table 2). For country i at wave t and follow-up horizon $h \in \{1, 2\}$ (+5 and +10 years), the four tax changes $Y_{i,t+h} = (Y_{1,i,t+h}, Y_{2,i,t+h}, Y_{3,i,t+h}, Y_{4,i,t+h})^\top \in \mathbb{R}^4$ are modelled jointly as:

$$Y_{i,t+h} = \underbrace{\mu\big(X_{i,t}, M_{i,t\to t+h}, \hat{z}(X_{i,t})\big)}_{\text{Baseline Trajectory at } Z_{i,t}=0} + \underbrace{\tau(W_{i,t})}_{\text{Causal Slope}} \cdot Z_{i,t} + \varepsilon_{i,t+h}, \quad \varepsilon_{i,t+h} \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(0, \Sigma). \tag{2}$$

To estimate the baseline trajectory $\mu(X_{i,t}, M_{i,t\to t+h}, \hat{z}(X_{i,t}))$, we use an ensemble of 75 decision trees, where each terminal leaf outputs a vector of four values corresponding to our four tax outcomes (Y_1 to Y_4). Whether these 75 decision trees are shallow or deep is not fixed in advance, but is learned directly from the data by the model during Markov chain Monte Carlo (MCMC) sampling. The baseline forest receives $X_{i,t}$ comprising 19 pre-treatment country characteristics measured at or before wave t (starting log real GDP per person in PPP dollars, starting total non-resource tax-to-GDP ratio L_1, starting direct income tax share L_3, starting goods and services tax share L_4, disposable income Gini inequality, life expectancy, gross tertiary and secondary enrolment ratios, total education spending as a share of GDP, working-age population share, log population, unemployment rate, natural resource rents, trade openness, lagged five-year GDP per person growth, World Bank income group, geographic region, OECD membership, and wave time index; starting direct income tax-to-GDP $L_2 = L_1 L_3/100$ is implicitly captured by tree splits on L_1 and L_3; see Table 2). In our primary specification (`M2a` at +5 years and `M2b` at +10 years), $\mu(\cdot)$ also receives four follow-up macroeconomic mediators $M_{i,t\to t+h}$ over $[t, t+h]$ (forward GDP per person growth, university enrolment expansion, income inequality changes, and life expectancy changes) to isolate the controlled direct fiscal effect $\tau^{\text{dir}}(W_{i,t})$ net of overall economic growth under sequential ignorability of $Z_{i,t}$ and $M_{i,t\to t+h}$ (Imai et al., 2010; Acharya et al., 2016; VanderWeele, 2016). Because holding follow-up GDP growth $M_{i,t\to t+h}$ fixed in a single stage could introduce collider bias if an unobserved 10-year economic boom simultaneously raises both GDP growth and tax receipts (slightly attenuating the estimated university funding slope), we treat `M2a/M2b` (+0.24 pp on Y_1 and +0.12 pp on Y_2 at +10 years) as a conservative lower bound alongside the unmediated total causal effect $\tau^{\text{tot}}(W_{i,t})$ omitting $M_{i,t\to t+h}$ entirely (`M1a/M1b`: +0.30 pp on Y_1 and +0.16 pp on Y_2 in Appendix Table 4) and two-stage sequential g-estimation (`R9`: +0.46 pp on Y_1 and +0.20 pp on Y_2 in Appendix Table 5; Acharya et al. 2016).

In standard Bayesian Causal Forests (Hahn et al., 2020), the treatment variable Z is binary and a propensity score $P(Z = 1 \mid X)$ is included inside $\mu(\cdot)$ to control for selection bias. However, in our approach the university funding variable $Z_{i,t} \in [-1.5, +1.5]$ is continuous rather than binary, so a binary propensity score no longer applies. Instead, following the partially linear and orthogonalised forest literature (Robinson, 1988; Chernozhukov et al., 2018; Athey et al., 2019), we estimate each country-wave's predicted continuous funding level $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ strictly from the 19 pre-treatment covariates $X_{i,t}$ (excluding post-treatment mediators $M_{i,t\to t+h}$) using a 500-tree random forest (`ranger`; Breiman 2001; Wright and Ziegler 2017) with 5-fold country-blocked cross-fitting, and we include $\hat{z}(X_{i,t})$ as a covariate inside $\mu(\cdot)$. Under conditional mean independence of treatment variation around $\hat{z}(X_{i,t})$ (a first-moment analogue of the generalised propensity score of Hirano and Imbens 2004), including $\hat{z}(X_{i,t})$ guards against regularisation-induced confounding (Hahn et al., 2020) and identifies $\tau(W_{i,t})$ from local funding deviations around what a country with similar history is predicted to spend. In Appendix D, we

confirm that adjusting for $\hat{z}(X_{i,t})$ reduces the mean absolute correlation between Z_1 and the 19 starting covariates by more than 60% (from 0.157 to 0.061).

To allow the fiscal slope per unit of university funding to vary across countries, we model $\tau(W_{i,t})$ using a second ensemble of 30 decision trees that split on the eight pre-treatment moderators $W_{i,t} = (W_{1,i,t}, \ldots, W_{8,i,t})^\top \subset X_{i,t}$: gross tertiary enrolment (W_1), log real GDP per person (W_2), gross secondary enrolment (W_3), starting direct income tax share (W_4), working-age population share (W_5), World Bank income group (W_6), OECD membership (W_7), and wave time index (W_8). Again, the exact tree structures and depths in $\tau(W_{i,t})$ are chosen by the data during MCMC sampling, but following Hahn et al. (2020) we place stronger regularisation priors on the τ tree shapes than on the μ tree shapes so that the treatment forest shrinks noisy subgroup differences toward the overall average slope (full hyperparameter details are provided in Appendix A). Because $Z_{i,t} \in [-1.5, +1.5]$ is centred at zero (the income-group median benchmark for Z_1, Z_3, Z_4, or zero change for Z_2), $\mu(X_{i,t}, M_{i,t\to t+h}, \hat{z}(X_{i,t}))$ directly represents the expected baseline tax trajectory when $Z_{i,t} = 0$, allowing us to adjust funding $Z_{i,t}$ up ($Z_{i,t} > 0$) or down ($Z_{i,t} < 0$) symmetrically around zero (Check `R16` in Appendix Table 5 confirms that shifting location by +1.5 onto $[0, 3.0]$ yields virtually identical slopes). All four tax outcomes (Y_1 to Y_4) are linked through the 4×4 residual covariance matrix Σ, which receives a conjugate Inverse-Wishart prior; whilst Equation (2) uses a working i.i.d. $\mathcal{N}_4(0, \Sigma)$ likelihood across country-waves, we verify that accounting for within-country serial correlation and overlapping two-wave horizons via country-clustered posterior standard errors (reported in Appendix Table 8), non-overlapping 10-year decadal waves (`R2`), and within-country fixed-effect demeaning (`R3a/R3b`) leaves our conclusions intact.

An important modelling question is why we feed all 19 starting covariates $X_{i,t}$ (plus the four follow-up growth controls $M_{i,t\to t+h}$) into the baseline forest $\mu(\cdot)$, while feeding the targeted subset of eight structural moderators $W_{i,t} \subset X_{i,t}$ into the treatment forest $\tau(\cdot)$ (see Table 2 in Appendix B for the side-by-side mapping of X and W). In Bayesian causal forest modelling (Hahn et al., 2020; McJames et al., 2025), separating prognostic covariates X from effect moderators W is recommended when the sample size is moderate ($N = 283$ to 371 country-waves). The 75 baseline trees in $\mu(X_{i,t}, M_{i,t\to t+h}, \hat{z}_{i,t})$ need all 19 starting variables and growth controls to absorb complex macroeconomic confounding, whereas restricting $\tau(W_{i,t})$ to the eight core institutional, educational, and demographic moderators prevents the 30 treatment trees from splitting on noisy short-term nuisance variables (such as trade openness or resource rents). However, one could reasonably argue that both parts of the model should be given the exact same set of covariates ($W_{i,t} = X_{i,t}$) so that the data alone decide which variables moderate the treatment slope. To test whether this choice affects our findings, we run a dedicated sensitivity check in Appendix Table 5 (Check `R15`) where $\mu(\cdot)$ and $\tau(\cdot)$ use the identical covariate matrix ($W_{i,t} = X_{i,t}$); doing so yields virtually identical 10-year posterior slopes (+0.33 pp on total tax/GDP Y_1, $P = 0.97$; +0.12 pp on direct income tax/GDP Y_2, $P = 0.85$; and +0.51 pp on the direct income tax share Y_3, $P = 0.91$).

The main advantage of the continuous-treatment MV-BCF framework is that it estimates smooth funding slopes, captures non-linear confounding and diminishing returns across country tiers, and shares information across correlated tax outcomes whilst providing full Bayesian posterior distributions. The main disadvantage is that tree ensembles rely on MCMC sampling and conditional ignorability assumptions rather than linear fixed-effect differencing, and assume locally linear funding slopes within terminal leaves over the central ±15 pp range $Z_{1,i,t} \in [-1.5, +1.5]$. All MV-BCF models are fitted in C++ for 2,000 MCMC iterations (750 burn-in, 1,250 retained posterior draws). We report convergence, residual, and out-of-sample holdout validation diagnostics in Section 5.5 and Appendices D and E.

5 Results

5.1 Continuous Funding Slopes After 5 and 10 Years

We begin by examining our primary question using funding measure Z_1. When a country increases its per-student university spending ratio by +10 percentage points of GDP per person ($\Delta Z_1 = +1.0$ unit), what happens to its four tax outcomes (Y_1 to Y_4) after 5 and 10 years once starting country conditions, predicted funding $\hat{z}(X_{i,t})$, and follow-up economic growth are controlled for? Figure 3 plots the posterior

mean slopes, 80% and 95% credible intervals, and posterior probabilities $P = \Pr(\tau > 0 \mid \text{data})$ across all four tax outcomes at +5 years ($N = 371$, 121 countries) and +10 years ($N = 283$, 112 countries).

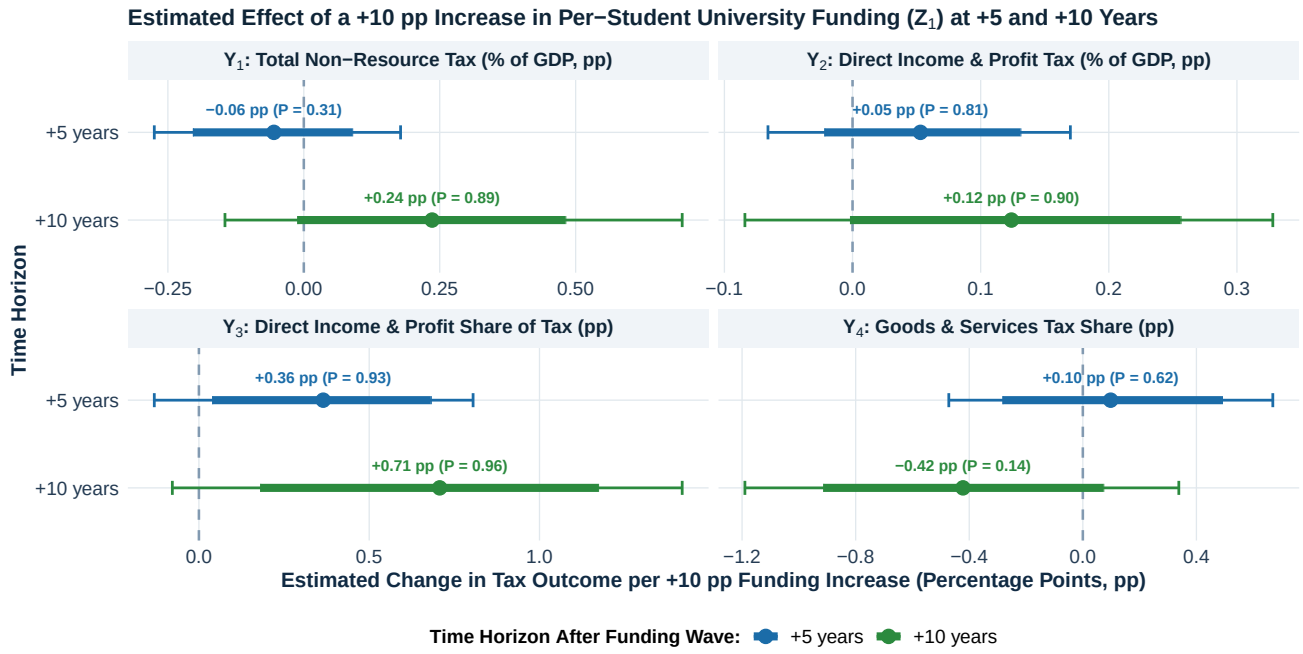


Figure 3: Posterior mean treatment slopes (in percentage points, pp, per +10 pp increase in per-student university spending Z_1) across the four jointly modelled tax outcomes (Y_1 to Y_4) at +5 years ($N = 371$, blue) and +10 years ($N = 283$, green). Thick horizontal bars show 80% Bayesian posterior credible intervals, thin horizontal lines show 95% credible intervals, and $P = \Pr(\tau > 0 \mid \text{data})$ denotes the posterior probability that the causal slope is greater than zero given the observed data.

Figure 3 reveals a clear maturation pattern across the four panels from Y_1 to Y_4, with the +5-year effect shown at the top of each panel and the +10-year effect shown below it. Starting in the top-left panel (Y_1, total non-resource tax revenue as a share of GDP), a +10 pp increase in per-student university spending (Z_1) has a slightly negative effect after 5 years at −0.06 pp of GDP ($P = 0.31$) while students are still enrolled, before rising to +0.24 pp of GDP after 10 years ($P = 0.89$, or +0.30 pp with $P = 0.95$ when follow-up growth mediators are omitted in Appendix Table 4). In the top-right panel (Y_2, direct income, profit, and capital gains tax revenue as a share of GDP), the slope is +0.05 pp of GDP after 5 years ($P = 0.81$) and more than doubles to +0.12 pp of GDP after 10 years ($P = 0.90$, or +0.16 pp with $P = 0.96$ for the total effect) as university cohorts graduate and advance into higher-salaried careers. In the bottom-left panel (Y_3, direct income and profit taxes as a share of total tax revenue), higher per-student funding produces a strong and sustained shift in the tax mix toward direct taxation, rising by +0.36 pp after 5 years ($P = 0.93$) and by +0.71 pp after 10 years ($P = 0.96$). Finally, in the bottom-right panel (Y_4, indirect goods and services taxes as a share of total tax revenue), the slope is flat after 5 years at +0.10 pp ($P = 0.62$) and shifts downward after 10 years by −0.42 pp ($P = 0.14$, or −0.65 pp with $P = 0.04$ for the total effect) as direct income taxation expands its share of the revenue pie.

To put these continuous slopes into concrete national currency terms, we hold each country's latest population and 2021 PPP GDP per person constant so that demographic growth does not inflate the calculation. At the 10-year mark, an annual direct income tax gain (Y_2) of +0.12 to +0.16 pp of GDP (and +0.24 to +0.30 pp in total non-resource tax Y_1) per +10 pp funding increase corresponds to roughly €0.23 billion to €0.30 billion per year in additional direct income tax receipts (and €0.44 billion to €0.55 billion per year in total tax) for a median-sized economy with a GDP of €183 billion. In a medium-sized high-income economy such as Ireland, the Year-10 annual gain from a +10 pp funding increase is €0.36 billion per year in direct income tax and €1.21 billion per year in total non-resource tax, and in a large economy such as the United Kingdom it is €2.52 billion per year in direct income tax and €7.29 billion per year in total non-resource tax. Because our data are averaged into five-year waves, a +10 pp increase in Z_1 represents an annual spending commitment (averaging 0.40% of GDP per year across below-median

countries, or €0.73 billion per year in a median economy and €2.37 billion per year in Ireland) sustained over the five-year wave while a student cohort is at university. Comparing a single year of tuition costs for the entire student body against a single year of tax receipts at Year 10 (when only the first two five-year graduate cohorts have entered the workforce) can make the annual funding cost look larger than the Year-10 annual return in high-enrolment countries, even though each cohort only requires university funding during its degree and then pays higher taxes across a 40-year working career; in Section 5.4 and Appendix F, we translate these 10-year causal slopes directly into a single-cohort return on investment across a 40-year career.

5.2 Comparing the Four Ways of Measuring University Funding (Z_1 to Z_4)

Whilst Figure 3 compared the +5-year and +10-year time horizons for our primary per-student spending ratio (Z_1), a natural follow-up question is how the +10-year fiscal slope changes when we unbundle the UNESCO ratio into its separate components. In Figure 4, we hold the time horizon fixed at +10 years and compare our four continuous university funding variables (Z_1 to Z_4), each estimated in its own four-outcome MV-BCF model. Specifically, we compare (1) the per-student spending ratio Z_1 (per +10 pp of GDP per person per student, $N = 283$, 112 countries), (2) observed within-country five-year funding changes Z_2 (per +10 pp five-year expansion, $N = 176$, 86 countries), (3) real 2021 PPP dollars per student Z_3 (per +25% relative logarithmic increase, removing GDP per person from the denominator, $N = 283$, 112 countries), and (4) the public university budget as a percentage of GDP Z_4 (per +0.25 pp of GDP, removing student headcount from the denominator, $N = 403$, 123 countries). Comparing Z_1 to Z_4 on the same 10-year horizon tests whether future tax gains are driven by real instructional resources per student or merely by aggregate budget size.

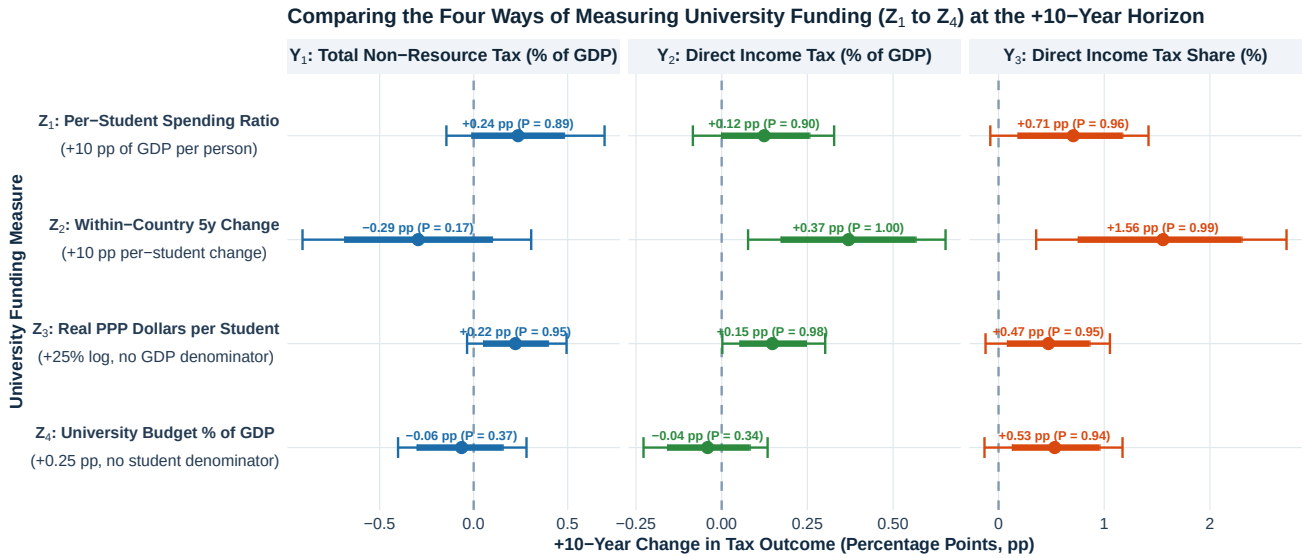


Figure 4: Comparison of +10-year posterior mean slopes (in percentage points, pp) across the four continuous university funding measures (Z_1 to Z_4, estimated in separate models) for total non-resource tax/GDP (Y_1), direct income tax/GDP (Y_2), and the direct income tax share of revenue (Y_3). Thick horizontal bars show 80% Bayesian credible intervals, thin lines show 95% credible intervals, and $P = \Pr(\tau > 0 \mid \text{data})$ denotes the posterior probability of a positive effect. Protecting real resources per student (Z_1, Z_2, and Z_3) raises 10-year direct income tax revenue, whereas expanding the aggregate budget without accounting for student numbers (Z_4) yields near-zero tax-to-GDP gains.

Figure 4 shows a simple pattern across the four ways of measuring university funding. The three measures that track resources *per student* (Z_1, Z_2, and Z_3) all produce positive 10-year gains in direct income tax revenue (Y_2) and in the direct income tax share (Y_3). Removing GDP from the denominator by measuring real PPP dollars per student (Z_3) raises 10-year total tax revenue (Y_1) by +0.22 pp ($P = 0.95$) and direct income tax revenue (Y_2) by +0.15 pp ($P = 0.98$). Focusing on five-year within-country funding increases (Z_2) produces the largest gains in direct income taxation (+0.37 pp on Y_2 and +1.56 pp on Y_3, $P \geq 0.99$), even though total tax-to-GDP (Y_1) dips slightly because reforming countries often

reduce trade and sales taxes at the same time (see Appendix C for the algebraic decomposition). By contrast, expanding the overall university budget without accounting for student numbers (Z_4) leaves 10-year tax-to-GDP slightly negative (-0.06 pp on Y_1, $P = 0.37$, and -0.04 pp on Y_2, $P = 0.34$, which holds on the overlapping $N = 272$ sample in Check `R17` of Appendix Table 5), confirming that funding must keep pace with enrolment to avoid institutional crowding (Bound et al., 2010; Deming and Walters, 2017; Chakrabarti et al., 2020).

5.3 Subgroup Moderation, Negative Scenarios, and Cohort Maturation Trajectories

How do the causal slopes per +10 pp of per-student funding (Z_1) vary across countries at different stages of university enrolment and economic development, and how does the model capture diminishing marginal returns? Figure 5 plots the +10-year subgroup posterior mean slopes and 95% credible intervals across starting university enrolment tiers and World Bank income groups for Y_1, Y_2, and Y_3 (with full numerical breakdowns, including our five-group OECD financing regime sub-model, reported in Appendix Table 3). As shown in Figure 5, the two fiscal dimensions vary in contrasting ways across development stages. On the tax-mix dimension (Y_3, right panel), the shift toward direct income and profit taxation rises as economies develop a formal corporate and salaried sector capable of hiring university graduates. The Y_3 slope is smallest in low-income economies (+0.16 pp, $P = 0.69$), averages +0.64 pp ($P = 0.91$) in elite systems with under 20% enrolment, rises to +0.74 pp ($P = 0.96$) in lower-middle-income economies, +0.75 pp ($P = 0.97$) in massifying systems with 20–50% enrolment, and +0.84 pp ($P = 0.98$) in upper-middle-income economies, and remains strongly positive at +0.68 to +0.73 pp ($P = 0.94$–0.95) in high-income and high-participation systems ($\geq 50\%$ enrolment, and +1.05 to +1.14 pp across all five OECD financing regimes). By contrast, on the aggregate tax-to-GDP dimensions (Y_1 and Y_2, left and middle panels), the treatment trees capture clear diminishing returns as university systems expand: the largest gains per +10 pp of funding occur in low-income (+0.48 pp on Y_1 and +0.24 pp on Y_2, $P = 0.91$), lower-middle-income (+0.32 pp and +0.16 pp, $P = 0.91$–0.93), and elite-enrolment (+0.34 pp and +0.17 pp, $P = 0.91$–0.93) economies, tapering off to +0.19 pp ($P = 0.82$) and +0.10 pp ($P = 0.81$) in high-income economies.

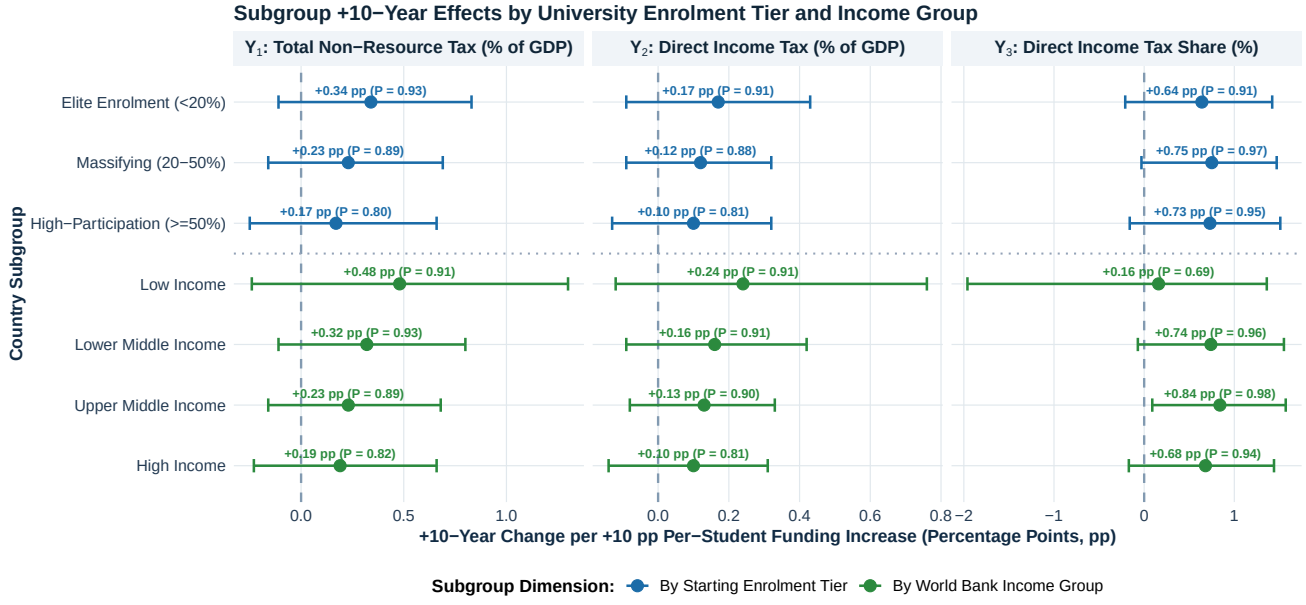


Figure 5: Subgroup +10-year posterior mean slopes (in percentage points, pp per +10 pp per-student funding increase Z_1) across starting university enrolment tiers and World Bank income groups for total non-resource tax/GDP (Y_1), direct income tax/GDP (Y_2), and the direct income tax share (Y_3). Horizontal bars show 95% Bayesian credible intervals, and $P = \Pr(\tau > 0 \mid \text{data})$ gives the posterior probability of a positive effect.

Both the historical data and our model estimates also highlight three specific scenarios where increasing university funding is associated with a *decrease* in the tax take. First, as annotated in Panel A of Figure 6, in the short run (+5 years) 243 out of 371 country-waves (65.5%) have negative point estimates

on total tax-to-GDP Y_1 (down to -0.96 pp of GDP per $+10$ pp funding increase, with an overall mean of -0.06 pp and -0.30 pp in low-income economies). This short-run dip reflects a temporary "student lock-in" and cohort-gestation effect: even when follow-up enrolment headcount is controlled for, higher per-student funding supports full-time study persistence and degree completion so that young adults spend more time in lecture theatres rather than working full time during those initial five years. Second, at the $+10$-year horizon in very low-income economies with tiny elite enrolment below 2% and extreme UNESCO ratios above 150% (such as Ethiopia in `1995-1999` at -0.94 pp and Malawi in `2010-2014` at -0.68 pp on Y_3), pouring additional subsidies into a tiny elite university sector without formal-sector job creation or protection against skilled emigration (Beine et al., 2008) fails to modernise the domestic income tax share. Third, when we fit a separate sub-model restricted exclusively to the 119 OECD country-waves where total tax revenue already exceeds 35% of GDP (Rows 8 to 12 of Appendix Table 3, where within-OECD Y_1 slopes average -0.14 to -0.17 pp per $+10$ pp increase, compared with $+0.23$ pp when evaluated inside the pooled $N = 283$ global model), expanding public university budgets within mature welfare states does not raise aggregate tax-to-GDP further even as the income tax share of revenue (Y_3) rises by $+1.05$ to $+1.14$ pp ($P = 0.93$–0.95).

To show how these effects evolve from the initial $+5$-year window to the $+10$-year window across income groups, Figure 6 plots the four World Bank income-group trajectory lines alongside the global posterior mean (dotted line) and the 80% and 95% posterior credible fan. In Panel A of Figure 6, total non-resource tax revenue as a share of GDP (Y_1) dips below zero across every income group at the $+5$-year mark (-0.30 pp in low-income, -0.04 pp in lower-middle-income, -0.01 pp in upper-middle-income, and -0.06 pp in high-income economies, with 65.5% of individual country-waves below zero) whilst students are still enrolled in university. By the $+10$-year mark in Panel A, Y_1 rebounds into positive territory across all four groups ($+0.48$ pp in low-income, $+0.32$ pp in lower-middle-income, $+0.23$ pp in upper-middle-income, and $+0.19$ pp in high-income economies). In Panel B of Figure 6, the direct income and profit tax share of revenue (Y_3) already rises at $+5$ years across all four groups ($+0.29$ to $+0.49$ pp) and then diverges sharply at $+10$ years. In lower-middle-income ($+0.74$ pp), upper-middle-income ($+0.84$ pp), and high-income ($+0.68$ pp) economies where formal salaried employment can absorb new graduates, the shift toward direct income taxation roughly doubles by Year 10, whereas in low-income economies it flattens out at $+0.16$ pp. In Appendix C, we confirm that this 10-year shift toward direct income taxation holds across 17 sensitivity checks (`R1` to `R17`), including within-country fixed-effect demeaning, non-overlapping decadal waves, pre-trend controls, sequential g-estimation, identical covariates in μ and τ ($W = X$), zero-centred treatment scaling, and competing Bayesian and frequentist estimators.

5.4 Cohort Return on Investment for Policy Makers

To answer whether increasing government spending on tertiary education provides a positive return on investment, we must compare the public cost of educating a cohort of students on the exact same basis as the taxes that cohort pays across its working life. In any calendar year, a country's university system holds all D year-groups of students simultaneously (for example, 1st-, 2nd-, 3rd-, and 4th-year undergraduates, or roughly 245,000 students in total in Ireland, corresponding to about 60,000 students per entering year-group). Because a single entering cohort makes up $1/D$ of the total university student body in any single year and studies for D years, the total multi-year cost of funding that one cohort across all D years of their degree (D years $\times\, 1/D$ of the student body) is algebraically identical to one year of the system-wide $+10$ pp university budget increase (€2.37 billion in Ireland, or one-fifth of a five-year wave). Once that cohort graduates and works across a 40-year career (ages 22 to 62, spanning eight five-year waves), our $+10$-year causal slope (τ_{10}, which measures the annual tax gain from a five-year wave of five cohorts) implies that each single cohort generates one-fifth of the Year-10 annual tax flow in every working year for 40 years. Consequently, under the benchmark accounting translation that the cohort sustains its Year-10 annual tax contribution across its working life (when tax receipts grow in line with GDP, $r = g$), the cumulative 40-year career tax output per cohort equals eight times the Year-10 annual flow (40 working years $\div$ 5 cohorts $= 8$; equivalently, multiplying both the cohort input and career output by 5 gives the five-year wave total with the exact same benefit-cost ratio and probability of a net positive return). Under alternative accounting assumptions, applying a net real treasury discount rate of

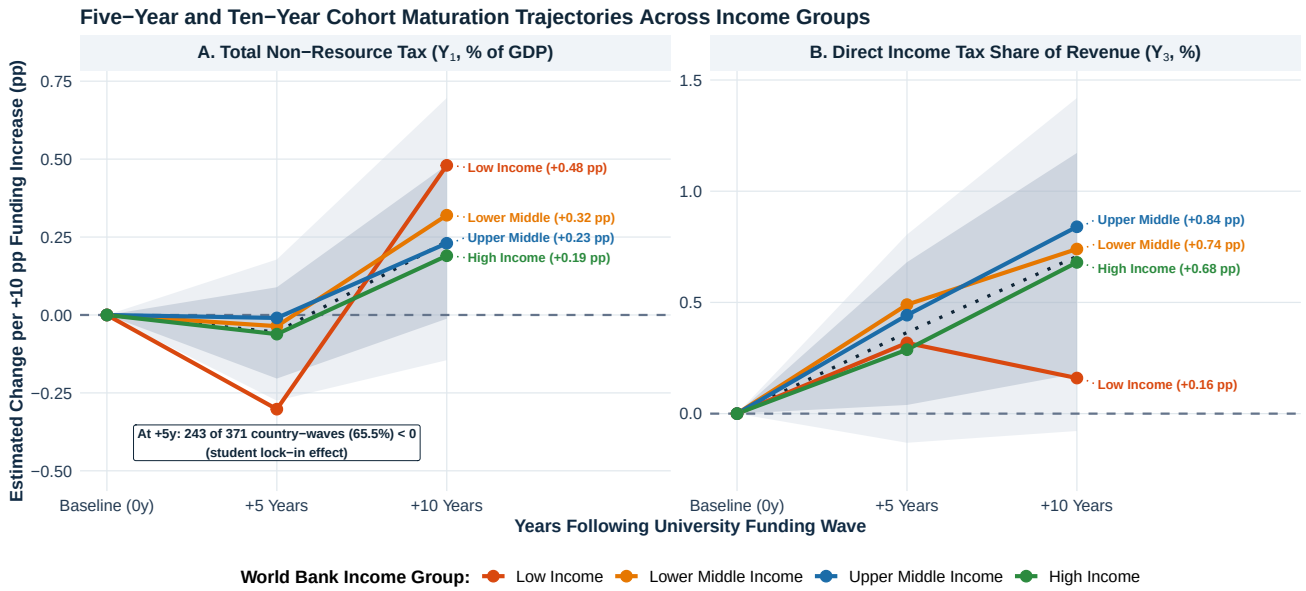


Figure 6: Five-year ($N = 371$) and 10-year ($N = 283$) cohort maturation trajectories of the causal slopes per +10 pp university spending increase (Z_1) across World Bank income groups for (A) total non-resource tax revenue as a percentage of GDP (Y_1, pp) and (B) the direct income and profit tax share of revenue (Y_3, pp). Coloured lines show income-group mean trajectories (with the callout in Panel A noting that 243 of 371 country-waves, or 65.5%, are negative at +5 years due to student lock-in), the dotted line marks the global posterior mean, and the shaded bands show the global 80% (dark grey) and 95% (light grey) posterior credible fan.

$r - g = 1.5\%$ per year across years 5 to 44 after university entry scales this 40-year multiplier from 8.0× to 5.7× in discounted present value, whilst assuming that two consecutive five-year waves of persistent spending (Z_1) have entered the labour force by Year 10 provides a conservative 4.0× lower bound and using the unmediated total effect (`M1b`) or five-year funding expansion impulse (Z_2) yields substantially higher career multipliers. We emphasise that this 40-year career calculation is inherently speculative because we are extrapolating our 10-year empirical estimates across four decades of a cohort's working life (during which graduating many cohorts could also gradually compress the graduate wage premium), so these figures should be interpreted as an illustrative order-of-magnitude benchmark rather than a guaranteed 40-year forecast.

When presenting these numbers to a national treasury, a policy maker faces a practical question: should the return on investment be judged using total non-resource tax revenue (Y_1) or direct income and profit tax revenue (Y_2)? For the bottom-line budget decision, the treasury should focus primarily on total non-resource tax revenue (Y_1), because universities are funded out of the general Exchequer pot and every euro of additional tax that flows back into the Exchequer counts toward paying back that outlay. When university graduates earn higher salaries, they do not only pay direct income and corporate profit taxes (Y_2). They and their employers also pay social insurance and payroll taxes (such as Pay Related Social Insurance in Ireland or National Insurance in the United Kingdom, which international accounts record separately from income tax), value-added and excise duties when graduates spend their higher take-home pay in the domestic economy, and property and business taxes generated by wider skilled economic activity. Looking only at Y_2 therefore leaves out more than two-thirds of the total tax revenue generated across a cohort's career. At the same time, because aggregate tax-to-GDP Y_1 is subject to broader macroeconomic noise (so out-of-sample cross-validation skill in Section 5.5 is strongest on Y_3 and Y_2, and the small $N = 119$ OECD-only sub-model shrinks Y_1 below zero in mature welfare states), policy makers should treat Y_1 (the full Exchequer return from the pooled global model) and Y_2 paired with Y_3 (the conservative direct income-tax floor) as joint upper and lower bounds.

Evaluating this cohort return on investment across the 70 countries with complete 2020–2024 macroeconomic, enrolment, and fiscal data (shown in Appendix F, Figure 10 and Table 10, and for all 70 countries in our interactive web application) reveals positive expected 40-year career returns across all income groups. In Ireland, where current university spending is 17.3% of GDP per person (USD

15,150 in 2021 PPP dollars), raising per-student funding by +10 pp requires a cohort degree input of €2.37 billion (0.43% of GDP). Across that cohort's 40-year working career, this €2.37 billion input is estimated to generate €+9.66 billion in total non-resource tax output Y_1 (95% CI [€ − 10.3B, € + 31.9B], benefit-cost ratio BCR = 4.08×, posterior probability of a net positive return $P_{\text{net}} = 0.77$; or €+6.90 billion, BCR = 2.91× when discounted at $r - g = 1.5\%$, and €+4.83 billion, BCR = 2.04× under the conservative 4× two-wave bound) and €+2.87 billion in direct income tax output Y_2 alone (95% CI [€ − 8.3B, € + 13.0B], BCR = 1.21×, $P_{\text{net}} = 0.54$), alongside a +0.73 pp shift toward direct income taxation ($P = 0.96$). Similarly, in the United Kingdom, a €11.55 billion cohort input (0.35% of GDP) generates an estimated €+58.33 billion in 40-year career total tax output Y_1 (95% CI [€ − 60.0B, € + 189.6B], BCR = 5.05×, $P_{\text{net}} = 0.79$; or BCR = 3.60× at a 1.5% net discount rate) and €+20.14 billion in direct income tax output Y_2 (95% CI [€ − 46.8B, € + 85.9B], BCR = 1.74×, $P_{\text{net}} = 0.60$). Across high-enrolment OECD economies, the expected career return on total tax Y_1 is typically 3× to 5× the cohort input ($P_{\text{net}} = 0.68$–0.81) while the conservative direct income-tax floor Y_2 is 0.6× to 2.9× ($P_{\text{net}} = 0.38$–0.68). In middle- and low-income economies with smaller university cohorts (such as Colombia, Malaysia, Mexico, Bangladesh, and Rwanda), raw benefit-cost ratios are much larger (3.9× to 41.4× on Y_1 and 1.8× to 25.5× on Y_2) because a +10 pp per-student funding boost costs a tiny fraction of national GDP when enrolment is small (0.07% of GDP in Rwanda and 0.19% in Bangladesh) while the macro slope $\tau_{10}(W_i)$ captures wider formalisation gains across GDP; scaling those developing-country returns proportionally by their tertiary enrolment rate relative to the high-income benchmark places their cohort benefit-cost ratios in the same 2.0× to 4.5× range as high-income economies.

5.5 MCMC Convergence and Out-of-Sample Country Validation

Because cross-country macro panels have modest sample sizes, it is important to check that our MCMC sampler converges cleanly and that the continuous MV-BCF model generalises accurately to countries that were excluded from the training run. Across all four tax outcomes at both +5 and +10 years (1,250 post-burn-in draws), split-chain $\hat{R}$ values lie between 0.999 and 1.030 (well below the 1.05 convergence threshold), effective sample sizes range from 65 to 120, and Monte Carlo standard errors are less than 13% of the posterior standard deviations (full convergence tables and trace plots are provided in Appendix D). Standardised posterior residuals show zero curvature across the funding range $Z_1 \in [-1.5, +1.5]$, with 94.0% to 95.8% of residuals falling within ±2 standard deviations and empirical 95% posterior predictive coverage between 95.8% and 97.9% (see Appendix E and Figure 9B).

To test how well the model predicts fiscal changes in unseen countries, Figure 7 reports 5-fold country-blocked out-of-sample cross-validation continuous ranked probability scores (CRPS, where lower values indicate more accurate probabilistic forecasts; Diebold and Mariano 1995; Gneiting and Raftery 2007) and root mean squared errors (RMSE) across all 112 countries ($N = 283$) for seven models. Averaged across all four outcomes, our continuous MV-BCF model achieves a mean out-of-sample CRPS of 2.33 pp (1.64 pp on Y_1, 0.85 pp on Y_2, 3.41 pp on Y_3, and 3.43 pp on Y_4, with 92.8% out-of-sample 95% interval coverage), outperforming historical training means (2.45 pp), ridge regression (2.52 pp), zero-change persistence (2.53 pp), and linear local projections (2.64 pp) whilst matching nonparametric random forests (2.35 pp) and single-forest BART (2.33 pp). Examining individual outcomes in Figure 7 shows why nonparametric tree models help most on tax composition: on noisy aggregate tax-to-GDP changes (Y_1), the unconditional historical training mean (1.60 pp CRPS, 3.05 pp RMSE) and random forest (1.59 pp CRPS) slightly edge out MV-BCF (1.64 pp CRPS), whereas on the direct income tax share (Y_3) MV-BCF (3.41 pp CRPS, 6.11 pp RMSE) and random forest (3.39 pp) substantially outperform the historical training mean (3.63 pp), ridge regression (3.71 pp), zero-change persistence (3.81 pp), and OLS local projections (3.82 pp). Furthermore, when trained exclusively on the 90 countries in Folds 1 to 4 and evaluated on the 22 locked holdout countries in Fold 5 ($N = 54$ waves), MV-BCF achieves 95% interval coverage of 88.9% to 96.3% and positive out-of-sample predicted treatment slopes ($\bar{\tau}_{\text{test}} = +0.44$ pp on Y_1, +0.21 pp on Y_2, and +0.72 pp on Y_3; see Appendix E, Table 9, and Figure 9).

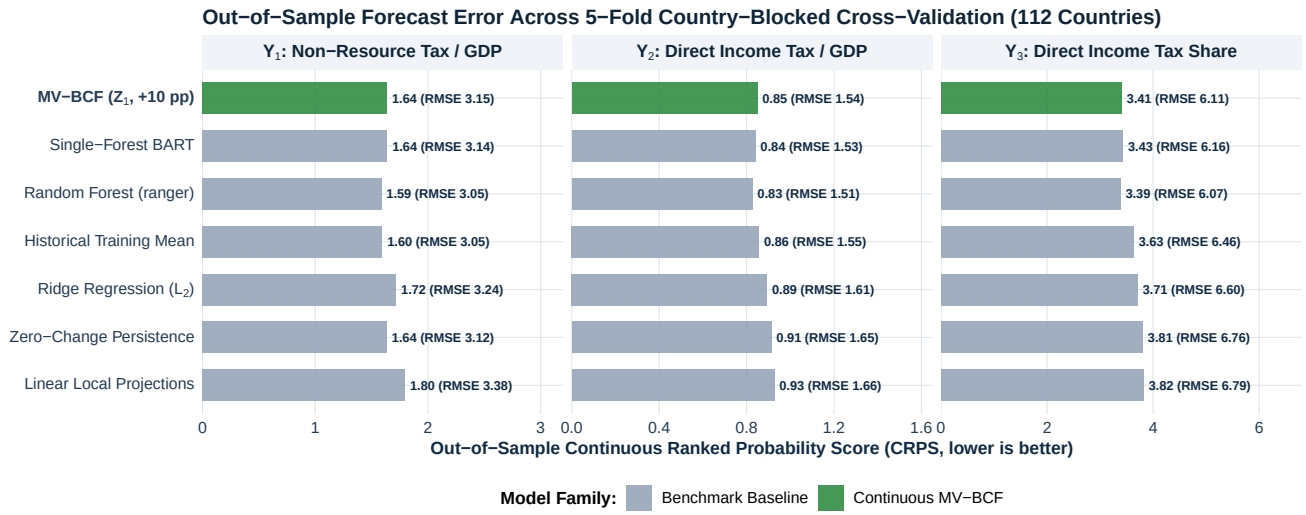


Figure 7: Out-of-sample 5-fold country-blocked cross-validation error across all 112 countries at the +10-year horizon ($N = 283$), comparing continuous MV-BCF (Z_1) against single-forest BART, random forests, historical training means, ridge regression (L_2), zero-change persistence, and linear local projections across Y_1 (total non-resource tax/GDP), Y_2 (direct income tax/GDP), and Y_3 (direct income tax share of revenue). Horizontal bars show mean out-of-sample Continuous Ranked Probability Score (CRPS, pp, lower is better) and text labels give CRPS and Root Mean Squared Error (RMSE, pp).

6 Conclusions and Discussion

6.1 Cohort Maturation Timelines and Multi-Cohort Payback Arithmetic

Our findings across 121 countries show that government spending on university education produces a measurable fiscal return, operating primarily by shifting the tax structure toward direct income, profit, and capital gains taxation over a 10-year horizon. Because our panel averages annual data into five-year waves, a university funding increase in wave t (years 0 to 4) is evaluated in wave $t+1$ (years 5 to 9) and wave $t+2$ (years 10 to 14). During the first five-year window, a +10 pp increase in per-student university funding (Z_1) immediately expands formal university payrolls whilst keeping a larger cohort of young adults in full-time study, which shifts the tax mix toward direct income tax ($Y_3 = +0.36$ pp, $P = 0.93$) without raising overall tax-to-GDP ($Y_1 = -0.06$ pp). By the 10-year window (years 10 to 14), the first two five-year university cohorts have completed their degrees and moved into salaried professional, managerial, and technical roles inside formal firms (Kleven et al., 2016; Valero and Van Reenen, 2019; Jensen, 2022), raising the direct income tax share (Y_3) by +0.71 pp ($P = 0.96$), direct income tax revenue (Y_2) by +0.12 to +0.37 pp of GDP, and total non-resource tax revenue (Y_1) by +0.22 to +0.30 pp of GDP.

For finance and education ministries, our comparison of the four funding definitions (Z_1 to Z_4) in Section 5.2 carries a clear practical lesson. Expanding the overall university budget (Z_4) or student numbers without protecting funding *per student* yields weak fiscal returns (−0.04 pp on Y_2). Countries that maintain or expand real resources per student avoid institutional crowding (Bound et al., 2010; Deming and Walters, 2017; Chakrabarti et al., 2020) and achieve much stronger 10-year gains in direct income tax receipts (+0.15 pp of GDP per +25% increase in real PPP dollars per student Z_3 and +0.37 pp of GDP per +10 pp within-country five-year expansion Z_2). Crucially, when comparing the upfront public cost of a per-student funding boost against the resulting tax return, policy makers must distinguish between a single-year snapshot at Year 10 and an illustrative lifecycle return across a student cohort's working career. Because a single university cohort only requires public tuition subsidies while enrolled in university, whereas after graduation that cohort works and pays taxes for roughly 40 years (eight five-year waves), extrapolating our 10-year estimates across a 40-year career provides a useful back-of-the-envelope lifecycle benchmark that explains why multi-decade microeconomic studies find high lifetime fiscal payback (Trostel, 2010; Hendren and Sprung-Keyser, 2020). Furthermore, our analysis focuses strictly on direct fiscal tax receipts and does not capture the many wider non-fiscal benefits

associated with university education - including lower crime rates, improved health and life expectancy, higher civic participation, intergenerational social mobility, and non-taxed research spillovers (Moretti, 2004; Trostel, 2010; Valero and Van Reenen, 2019; Chetty et al., 2020) - meaning our tax return estimates represent a conservative lower bound on the total social return to higher education investment.

6.2 Limitations and Future Research

Several caveats apply when interpreting observational cross-country macro panels, and we highlight five main limitations of our study:

1. *Reverse causality and fiscal-capacity feedback:* A central concern in macro-fiscal studies is that growing tax capacity enables governments to spend more on higher education rather than the other way round (Baumol, 1967; World Bank and UNESCO, 2024; World Bank, 2024). We adjust for starting tax-to-GDP, starting income and goods tax shares, lagged GDP growth, predicted continuous funding $\hat{z}(X_{i,t})$, and explicit pre-treatment fiscal trends (Appendix Table 5). Even so, observational conditioning cannot completely rule out multi-decade fiscal feedback loops.

2. *Unobserved concurrent policy reforms:* Our causal estimates rely on conditional ignorability given the 19 baseline characteristics and four follow-up growth controls (Imai et al., 2010; Acharya et al., 2016). If a reformist government simultaneously modernises its electronic tax administration (Pomeranz and Vila-Belda, 2019; Okunogbe and Pouliquen, 2022) and increases university funding within the same five-year wave, part of the estimated slope could reflect that concurrent tax reform.

3. *Skilled emigration (brain drain):* In low- and lower-middle-income countries, a substantial share of university graduates emigrate to higher-income economies (Beine et al., 2008). Because emigrants pay direct income taxes in their destination country whilst sending untaxed remittances home (which are spent on domestic goods subject to VAT), high skilled emigration can dampen domestic direct income tax returns in smaller developing nations.

4. *Central government reporting, five-year wave coarseness, and overlapping horizons:* World Bank fiscal series primarily record central government revenue, which can understate state or provincial income taxes in federal countries. In addition, averaging annual data into five-year waves smooths out exact year-to-year budgetary timing, and consecutive 10-year forward differences ($t \to t+2$ and $t+1 \to t+3$) share a five-year window, which we verify via non-overlapping decadal waves (`R2`), country-clustered posterior probabilities (P_{clu}), and country-clustered frequentist causal forests (`B7`).

5. *Local linear slopes, diminishing returns, and career extrapolation assumptions:* Modelling university funding as a zero-centred continuous variable $Z_1 \in [-1.5, +1.5]$ avoids the artificial cliff-edge of a binary cutoff, so a country sitting just 0.31 pp below its income-group median (such as New Zealand at 26.1% vs the 26.4% high-income median) has a tiny gap cost (€0.04 billion per year) and a proportionally tiny gap return (€0.004 billion per year). As shown in Section 5.4 and Appendix F (Figure 10 and Table 10), comparing a single entering cohort's multi-year degree input cost against its cumulative 40-year career tax output resolves the apparent mismatch between annual system-wide tuition costs and single-year Year-10 tax flows in high-enrolment countries such as Ireland (€2.37 billion cohort input vs €9.66 billion career total tax output on Y_1, BCR $= 4.08\times$, $P_{\text{net}} = 0.77$, and €2.87 billion on Y_2, BCR $= 1.21\times$, $P_{\text{net}} = 0.54$). Even so, readers should bear in mind that our empirical panel directly observes outcomes up to the +10-year horizon, so multiplying the per-cohort Year-10 annual flow by 40 working years is a speculative cohort-accounting projection (which scales from 8.0× undiscounted to 5.7× under a 1.5% net real discount rate, or 4.0× under a conservative two-wave level attribution, and abstracts from long-run general-equilibrium compression of the graduate wage premium), and country-level 95% credible intervals remain wide. Finally, because Z_1 is winsorised at ±15 pp around each income group's median, our slopes measure local marginal returns over a realistic policy window rather than unlimited linear extrapolation.

These limitations point to several promising avenues for future research. First, whilst this paper has focussed on tertiary education because university cohorts enter the labour market within a 5-to-10-year window, extending continuous-treatment multivariate Bayesian causal forests to 15-to-25-year horizons would allow researchers to compare government spending on early childhood, primary, and secondary schooling (Hanushek and Woessmann, 2012; Hendren and Sprung-Keyser, 2020; Angrist et al., 2021) directly against university funding. Second, incorporating bilateral skilled migration data (Beine et al., 2008) directly into the treatment-effect moderators W would quantify how brain drain alters the fiscal return to public university subsidies in developing countries. Third, linking university funding cohorts to administrative employer-employee tax registers within individual countries would make it possible to trace graduate earnings and corporate tax payments over a full 40-year career. To help policy makers and researchers explore our dataset and country-level estimates interactively, we provide a dedicated policy-maker guide in Appendix F alongside an interactive web application in the project repository.

Declarations and Statements

Dedication. This paper is dedicated to the memory of Robert Christopher Parnell.

Author Contributions (CRediT). Andrew C. Parnell: Conceptualisation, Data curation, Formal analysis, Methodology, Software, Visualisation, Writing – original draft, Writing – review and editing.

Data Availability Statement. All raw and processed datasets analysed in this study are derived from publicly available international databases (World Bank WDI, UNESCO UIS, IMF, WGI, SWIID, and OECD) licensed under CC-BY-4.0. The complete dataset, SHA-256 checksums, and provenance documentation are openly available in the project repository at `https://github.com/andrewcparnell/education-tax-bcf`.

Code Availability Statement. All R and C++ (`RcppArmadillo`) source code, data-cleaning scripts, MCMC samplers, diagnostic scripts, interactive exploration app, automated pipeline tests, and figure generation pipelines are openly available under the MIT License at `https://github.com/andrewcparnell/education-tax-bcf`. The entire analysis and manuscript can be reproduced end-to-end via `bash run_all.sh`.

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Declaration of Generative AI and AI-Assisted Technologies. During the preparation of this work, the author used large language models (Anthropic Claude and Google Gemini within the Antigravity agentic coding environment) to assist with assembling the reproducible R and C++ data and MCMC pipeline, running multi-agent adversarial peer-review checks, building the companion interactive web application, and editing the manuscript text. All AI-generated code, statistical outputs, citations, and text were inspected, verified, and revised by the author. A complete stage-by-stage audit of the 840 LLM queries, token volumes, and estimated electrical energy and carbon footprint is provided in Appendix G (Table 11). The author takes full responsibility for the integrity and content of this publication.

Ethics and Conflict of Interest. This study analyses aggregate sovereign macroeconomic statistics and involves no human subjects or personally identifiable information, so institutional ethical approval is not required. As disclosed in Section 2, the author co-developed the Multivariate Bayesian Causal Forest methodology (McJames et al., 2025) evaluated alongside competing estimators in this study, and declares no financial or other competing interests.

A Mathematical Specification of the Continuous-Treatment MV-BCF Sampler

Let $Y_{i,t+h} = \left(Y_{1,i,t+h}, Y_{2,i,t+h}, Y_{3,i,t+h}, Y_{4,i,t+h}\right)^\top \in \mathbb{R}^4$ denote the vector of four disaggregated percentage-point fiscal changes for country i over horizon $h \in \{1, 2\}$ (+5 and +10 years), and let $Z_{i,t} \in [-1.5, +1.5]$ denote the zero-centred continuous university funding variable. Following Hahn et al. (2020) and McJames et al. (2025), the continuous-treatment Multivariate Bayesian Causal Forest (MV-BCF) models $Y_{i,t+h}$ as:

$$Y_{i,t+h} = \sum_{m=1}^{M_\mu} g_\mu\left(X_{i,t}, M_{i,t\to t+h}, \hat{z}_{i,t}; T_m^\mu, M_m^\mu\right) + Z_{i,t} \sum_{l=1}^{M_\tau} g_\tau\left(W_{i,t}; T_l^\tau, M_l^\tau\right) + \varepsilon_{i,t+h}, \qquad \varepsilon_{i,t+h} \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(0, \Sigma), \tag{3}$$

where $\hat{z}_{i,t} = \hat{\mathbb{E}}[Z_{i,t} \mid X_{i,t}]$ is the random forest predicted funding level given the 19 pre-treatment covariates $X_{i,t}$. The prognostic forest uses $M_\mu = 75$ decision trees with depth-penalty prior hyperparameters ($\alpha_\mu = 0.95, \beta_\mu = 2.0$), and the treatment-effect forest uses $M_\tau = 30$ decision trees with stronger regularisation priors on tree depth ($\alpha_\tau = 0.25, \beta_\tau = 3.0$) (Hahn et al., 2020). Each prognostic leaf vector $\mu_{m,b} \in \mathbb{R}^4$ and treatment slope leaf vector $\tau_{l,b} \in \mathbb{R}^4$ receives an independent diagonal multivariate Gaussian prior $\mu_{m,b} \sim \mathcal{N}_4(0, \Sigma_\mu)$ and $\tau_{l,b} \sim \mathcal{N}_4(0, \Sigma_\tau)$ calibrated to the empirical outcome variances $s_d^2 = \widehat{\text{Var}}(Y_{d,i,t+h})$ ($d = 1, \ldots, 4$) and the treatment range $\Delta_Z = \max(\max_i Z_{i,t} - \min_i Z_{i,t}, 10^{-4})$:

$$\Sigma_\mu = \text{diag}\left(\frac{s_1^2}{M_\mu}, \ldots, \frac{s_4^2}{M_\mu}\right), \qquad \Sigma_\tau = \text{diag}\left(\frac{s_1^2}{M_\tau \Delta_Z^2}, \ldots, \frac{s_4^2}{M_\tau \Delta_Z^2}\right). \tag{4}$$

Let $R_i^{(l)} = Y_{i,t+h} - \sum_{m=1}^{M_\mu} g_\mu(X_{i,t}, M_{i,t\to t+h}, \hat{z}_{i,t}; T_m^\mu, M_m^\mu) - Z_{i,t} \sum_{k \neq l} g_\tau(W_{i,t}; T_k^\tau, M_k^\tau)$ denote the four-dimensional partial residual for treatment tree l (and analogously for prognostic tree m). Because $Z_{i,t}$ multiplies $\tau_{l,b}$ linearly inside leaf b, the conjugate treatment leaf posterior is exact multivariate Gaussian with precision $\Omega_{\tau,b}^{-1} = \Sigma_\tau^{-1} + (\sum_{i \in b} Z_{i,t}^2)\Sigma^{-1}$ and mean $\Omega_{\tau,b}\Sigma^{-1} \sum_{i \in b} Z_{i,t} R_i^{(l)}$ (with $Z_{i,t} \equiv 1$ and Σ_μ^{-1} for prognostic leaves). During reversible-jump Metropolis-Hastings tree updates (Chipman et al., 2010) with equal proposal probabilities $p_{\text{grow}} = p_{\text{prune}} = 0.5$, a `GROW` move on a terminal node η at depth d_η (with splitting probability $p_{\text{split}}(\eta) = \alpha(1 + d_\eta)^{-\beta}$ and minimum node size $n_{\min} = 3$) is accepted with probability $\min(1, r_{\text{grow}})$:

$$r_{\text{grow}} = \frac{p_{\text{prune}}\, b(T)}{p_{\text{grow}}\, w(T^*)} \cdot \frac{p_{\text{split}}(\eta)\left(1 - p_{\text{split}}(\eta_L)\right)\left(1 - p_{\text{split}}(\eta_R)\right)}{1 - p_{\text{split}}(\eta)} \cdot \frac{L(R \mid T^*, \Sigma)}{L(R \mid T, \Sigma)}, \tag{5}$$

where $b(T)$ is the number of terminal leaves in T and $w(T^*)$ is the number of singly internal (`nog`) parent nodes whose two children are both terminal leaves in the proposed tree T^* (with the inverse ratio used for `PRUNE` moves). Given full residuals $E_i = Y_{i,t+h} - \mu_i - \tau_i Z_{i,t}$, the cross-outcome covariance matrix Σ receives a conjugate Inverse-Wishart prior $\mathcal{IW}(\nu_0, \Psi_0)$ with $\nu_0 = D + 2 = 6$ and scale matrix $\Psi_0 = \text{diag}(s_1^2, s_2^2, s_3^2, s_4^2)$, and is updated at each Gibbs iteration via $\Sigma \mid E \sim \mathcal{IW}(\nu_0 + N, \Psi_0 + \sum_{i=1}^N E_i E_i^\top)$.

B Data Dictionary for Outcomes Y, Funding Measures Z, Baseline Controls X, Moderators W, and Mediators M

Table 2 provides the complete data dictionary for all variables entering the continuous-treatment Multivariate Bayesian Causal Forest model in Equation (2). Specifically, it lists the four jointly modelled

forward tax change outcomes (Y_1 to Y_4), the four continuous university funding measures (Z_1 to Z_4, each zero-centred on $[-1.5, +1.5]$ and tested in turn), the 19 pre-treatment country covariates (X_1 to X_{19}) that enter both the random forest model for $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ and the 75-tree baseline prognostic forest $\mu(\cdot)$, the eight pre-treatment moderators (W_1 to $W_8 \subset X$) that enter the 30-tree treatment-effect forest $\tau(W_{i,t})$, and the four follow-up macroeconomic growth mediators (M_1 to M_4) included inside $\mu(\cdot)$ in the mediator-removed direct-effect specifications (`M2a/M2b`). Full variable ranges and missingness statistics across all 62 raw and constructed columns are saved in `data/processed/data_dictionary.csv`.

Table 2: Complete data dictionary for the four tax change outcomes (Y_1 to Y_4), the four zero-centred continuous university funding measures (Z_1 to $Z_4 \in [-1.5, +1.5]$), the 19 pre-treatment baseline controls (X_1 to X_{19}), the eight treatment-effect moderators (W_1 to $W_8 \subset X$), and the four follow-up macroeconomic growth mediators (M_1 to M_4).

Symbol	Code Name	Model Role	Definition & Scaling Units	Primary Source
Y_1	d_t{1,2}_nonres_tax_gdp_pp	Y Outcome 1 (+5y, +10y)	Δ domestic non-resource tax revenue (pp of GDP)	IMF GFS / WDI / GRD
Y_2	d_t{1,2}_inc_tax_gdp_pp	Y Outcome 2 (+5y, +10y)	Δ direct income, profit & capital gains tax (pp of GDP)	IMF GFS / WDI
Y_3	d_t{1,2}_inc_tax_share_pp	Y Outcome 3 (+5y, +10y)	Δ direct income & profit share of tax revenue (pp)	WDI (GC.TAX.YPKG.ZS)
Y_4	d_t{1,2}_goods_tax_share_pp	Y Outcome 4 (+5y, +10y)	Δ goods & services (VAT/sales) share of tax revenue (pp)	WDI (GC.TAX.GSRV.RV.ZS)
Z_1	Z_cont_10pp	Primary Funding Z_1	Per-student spend (% GDP/person) centred on group median (per +10 pp)	UNESCO (SE.XPD.TERT.PC.ZS)
Z_2	Z_cont_delta_10pp	Funding Change Z_2	Within-country 5-year change in per-student ratio (per +10 pp)	UNESCO (SE.XPD.TERT.PC.ZS)
Z_3	Z_cont_real_ppp_log25pct	Real PPP Funding Z_3	Log real 2021 PPP $ per student centred on group median (per +25%)	UNESCO & WDI PPP GDP
Z_4	Z_cont_budget_gdp_025pp	Total Budget Z_4	Public tertiary budget (% of GDP) centred on group median (per +0.25 pp)	UNESCO (SE.XPD.TOTL.GD.ZS)
X_1 (W_1)	gross_tert_enrolment_pct	Control X_1 & Moderator W_1	Baseline gross tertiary enrolment ratio (% of age cohort)	UNESCO (SE.TER.ENRR)
X_2 (W_2)	log_gdp_pc_ppp	Control X_2 & Moderator W_2	Log baseline real GDP per person (constant 2021 PPP USD)	WDI (NY.GDP.PCAP.PP.KD)
X_3 (W_3)	gross_seco_enrolment_pct	Control X_3 & Moderator W_3	Baseline gross secondary school enrolment ratio (% of cohort)	UNESCO (SE.SEC.ENRR)
X_4 (W_4)	income_profit_tax_share_rev_pct	Control X_4 & Moderator W_4	Baseline direct income, profit & capital gains share of tax (L_3, %)	WDI (GC.TAX.YPKG.ZS)
X_5 (W_5)	working_age_pop_15_64_pct	Control X_5 & Moderator W_5	Baseline share of population aged 15 to 64 (%)	WDI (SP.POP.1564.TO.ZS)
X_6 (W_6)	income_group	Control X_6 & Moderator W_6	World Bank income group tier (1 = Low to 4 = High income)	World Bank Metadata
X_7 (W_7)	is_oecd	Control X_7 & Moderator W_7	Indicator for OECD member country (0 or 1)	OECD / World Bank
X_8 (W_8)	wave_index	Control X_8 & Moderator W_8	Five-year wave time index (1 = 1985–1989 to 8 = 2020–2024)	Constructed (5y windows)
X_9	non_resource_tax_gdp_pct	Baseline Control X_9	Baseline domestic non-resource tax revenue (L_1, % of GDP)	IMF GFS / WDI / GRD
X_{10}	goods_services_tax_share_rev_pct	Baseline Control X_{10}	Baseline goods & services tax share of revenue (L_4, %)	WDI (GC.TAX.GSRV.RV.ZS)
X_{11}	edu_spend_total_gdp_pct	Baseline Control X_{11}	Baseline total public education spending across all levels (% GDP)	UNESCO (SE.XPD.TOTL.GD.ZS)
X_{12}	gini_disp_harmonised	Baseline Control X_{12}	Baseline harmonised disposable income Gini index (0–100)	SWIID v9.6 (Solt 2020)
X_{13}	life_expectancy_birth_yrs	Baseline Control X_{13}	Baseline life expectancy at birth (years)	WDI (SP.DYN.LE00.IN)
X_{14}	log_population	Baseline Control X_{14}	Natural log of baseline total national population	WDI (SP.POP.TOTL)
X_{15}	unemployment_rate_pct	Baseline Control X_{15}	Baseline total unemployment rate (% of labour force)	WDI (SL.UEM.TOTL.ZS)
X_{16}	natural_resource_rents_gdp_pct	Baseline Control X_{16}	Baseline total natural resource rents (% of GDP)	WDI (NY.GDP.TOTL.RT.ZS)
X_{17}	trade_openness_gdp_pct	Baseline Control X_{17}	Baseline sum of exports and imports of goods & services (% GDP)	WDI (NE.TRD.GNFS.ZS)
X_{18}	lag_gdp_pc_growth_pct	Baseline Control X_{18}	Lagged (wave $t-1$) annual real GDP per person growth (%)	WDI (NY.GDP.PCAP.KD.ZG)
X_{19}	region	Baseline Control X_{19}	World Bank geographic region (7 global regions)	World Bank Metadata
M_1	d_t{1,2}_log_gdp_pc_pct	Mediator M_1 (+5y, +10y)	Follow-up real GDP per person PPP log growth over $[t, t+h]$ (%)	WDI (NY.GDP.PCAP.PP.KD)
M_2	d_t{1,2}_tert_enrol_pp	Mediator M_2 (+5y, +10y)	Follow-up change in gross tertiary enrolment over $[t, t+h]$ (pp)	UNESCO (SE.TER.ENRR)
M_3	d_t{1,2}_gini_disp	Mediator M_3 (+5y, +10y)	Follow-up change in disposable income Gini over $[t, t+h]$ (pts)	SWIID v9.6 (Solt 2020)
M_4	d_t{1,2}_life_exp_yrs	Mediator M_4 (+5y, +10y)	Follow-up change in life expectancy at birth over $[t, t+h]$ (years)	WDI (SP.DYN.LE00.IN)

C Subgroup Breakdowns, Method Benchmarks, and Robustness Checks

In this appendix we present and interpret the detailed subgroup slopes, competing estimator benchmarks, sensitivity checks, and country-level cost-return calculations referenced in Section 5. Table 3 reports the +10-year subgroup posterior mean slopes and 95% credible intervals per +10 pp funding increase (Z_1) across starting university enrolment tiers ($N = 244$ strictly observed enrolment rows out of 283; the remaining 39 waves receive training-median enrolment in the tree splits and have mean slopes of +0.24 pp on Y_1 and +0.70 pp on Y_3), all four World Bank income groups ($N = 283$), and the five OECD financing regimes ($N = 119$ OECD country-waves). Rows 1 to 7 of Table 3 aggregate the country-wave posterior draws $\tau(W_{i,t})$ directly from our primary $N = 283$ global model (`M2b`), whereas Rows 8 to 12 come from a dedicated OECD-only sub-model ($N = 119$, 37 OECD countries, $M_\mu = 50, M_\tau = 25$) that adds the OECD public tertiary funding share to $X_{i,t}$ and $W_{i,t}$. When the global $N = 283$ model is averaged across the 119 OECD country-waves, it predicts +0.23 pp on Y_1, +0.08 pp on Y_2, and +0.75 pp on Y_3; when the sample is restricted solely to the 119 high-tax OECD waves in Rows 8 to 12, the strong regularisation prior ($\alpha_\tau = 0.25, \beta_\tau = 3.0$) shrinks all five OECD tuition regimes toward a common within-OECD slope of −0.14 to −0.17 pp on Y_1, +0.04 to +0.09 pp on Y_2, and +1.05 to +1.14 pp ($P = 0.93$–0.95) on Y_3.

Table 4 compares our primary joint four-outcome continuous MV-BCF model at the +10-year horizon ($N = 283$, Z_1) against a specification that omits follow-up macroeconomic growth controls (Total Effect `M1b`), three Bayesian tree ablations (`B1` separate univariate BCFs, `B2` MV-BCF without predicted funding $\hat{z}(X)$, and `B3` single-forest BART), and four frequentist estimators (`B4` linear local projections, Jordà 2005; `B5` country-and-wave two-way fixed effects; `B6` 5-fold country-blocked cross-fitted double machine

Table 3: Subgroup +10-year posterior mean continuous slopes (in percentage points, pp per +10 pp university spending increase Z_1, with 95% CIs on the second line of each cell and posterior probabilities $P = \Pr(\tau > 0 \mid \text{data})$) across starting university enrolment tiers ($N = 244$ strictly observed rows from the $N = 283$ global model), World Bank income groups ($N = 283$ global model), and OECD financing regimes ($N = 119$ separate OECD sub-model). Subgroups with $N \leq 20$ are small-sample exploratory strata.

Subgroup (+10-Year Horizon, Z_1)	**Waves (Countries)**	**Total Tax / GDP (Y_1, pp)**	**Direct Income Tax / GDP (Y_2, pp)**	**Direct Income Tax Share (Y_3, pp)**
1. Elite Enrolment ($< 20\%$)	74 (43)	$+0.34$ ($P = 0.93$) $[-0.11, +0.83]$	$+0.17$ ($P = 0.91$) $[-0.09, +0.43]$	$+0.64$ ($P = 0.91$) $[-0.21, +1.42]$
2. Massifying (20–50%)	73 (42)	$+0.23$ ($P = 0.89$) $[-0.16, +0.69]$	$+0.12$ ($P = 0.88$) $[-0.09, +0.32]$	$+0.75$ ($P = 0.97$) $[-0.03, +1.47]$
3. High-Participation ($\geq 50\%$)	97 (44)	$+0.17$ ($P = 0.80$) $[-0.25, +0.66]$	$+0.10$ ($P = 0.81$) $[-0.13, +0.32]$	$+0.73$ ($P = 0.95$) $[-0.16, +1.51]$
4. Low Income (exploratory)	15 (9)	$+0.48$ ($P = 0.91$) $[-0.24, +1.30]$	$+0.24$ ($P = 0.91$) $[-0.12, +0.76]$	$+0.16$ ($P = 0.69$) $[-1.96, +1.36]$
5. Lower Middle Income	47 (21)	$+0.32$ ($P = 0.93$) $[-0.11, +0.80]$	$+0.16$ ($P = 0.91$) $[-0.09, +0.42]$	$+0.74$ ($P = 0.96$) $[-0.07, +1.55]$
6. Upper Middle Income	76 (35)	$+0.23$ ($P = 0.89$) $[-0.16, +0.68]$	$+0.13$ ($P = 0.90$) $[-0.08, +0.33]$	$+0.84$ ($P = 0.98$) $[+0.09, +1.57]$
7. High Income	145 (47)	$+0.19$ ($P = 0.82$) $[-0.23, +0.66]$	$+0.10$ ($P = 0.81$) $[-0.14, +0.31]$	$+0.68$ ($P = 0.94$) $[-0.17, +1.44]$
8. OECD Nordic High-Public	20 (5)	-0.17 ($P = 0.23$) $[-0.62, +0.38]$	$+0.04$ ($P = 0.57$) $[-0.34, +0.49]$	$+1.13$ ($P = 0.94$) $[-0.18, +2.51]$
9. OECD Low-Tuition Public	46 (15)	-0.15 ($P = 0.26$) $[-0.58, +0.36]$	$+0.05$ ($P = 0.58$) $[-0.33, +0.47]$	$+1.05$ ($P = 0.93$) $[-0.26, +2.38]$
10. OECD Hybrid Public-Fee	29 (8)	-0.15 ($P = 0.25$) $[-0.60, +0.35]$	$+0.04$ ($P = 0.56$) $[-0.33, +0.45]$	$+1.11$ ($P = 0.95$) $[-0.17, +2.45]$
11. OECD High-Tuition High-Sub.	11 (5)	-0.16 ($P = 0.24$) $[-0.63, +0.39]$	$+0.05$ ($P = 0.57$) $[-0.34, +0.50]$	$+1.14$ ($P = 0.94$) $[-0.16, +2.55]$
12. OECD High-Tuition Low-Sub.	13 (4)	-0.14 ($P = 0.27$) $[-0.59, +0.37]$	$+0.09$ ($P = 0.64$) $[-0.31, +0.49]$	$+1.09$ ($P = 0.94$) $[-0.30, +2.46]$

learning, Chernozhukov et al. 2018; and B7 a frequentist nonparametric orthogonal causal forest / R-learner fitted on $W_{i,t}$ using 500-tree random forests with country-clustered standard errors, Athey et al. 2019). Three methodological lessons stand out from Table 4. First, omitting follow-up growth controls strengthens the 10-year slopes on total non-resource tax/GDP ($Y_1 = +0.30$ pp, $P = 0.95$ vs $+0.24$ pp, $P = 0.89$) and direct income tax/GDP ($Y_2 = +0.16$ pp, $P = 0.96$ vs $+0.12$ pp, $P = 0.90$) whilst preserving a positive shift in the direct income tax share ($Y_3 = +0.54$ pp, $P = 0.92$ vs $+0.71$ pp, $P = 0.96$). Second, amongst the Bayesian tree models (B1–B3), univariate BCFs (B1) and MV-BCF without $\hat{z}(X)$ (B2) yield positive slopes closely aligned with our primary model (with the joint covariance matrix Σ in our primary model improving posterior precision on Y_3: SD = 0.385 vs 0.462 pp), whereas single-forest BART (B3) shrinks the treatment slopes toward zero ($+0.03$ pp on Y_2 and $+0.34$ pp on Y_3), illustrating regularisation-induced confounding when μ and τ do not receive separate priors (Hahn et al., 2020). Third, across the frequentist estimators (B4–B7), the nonparametric orthogonal causal forest (B7) closely matches our Bayesian MV-BCF slopes ($+0.25$ pp on Y_1, $P_{\text{freq}} = 0.92$; $+0.11$ pp on Y_2, $P_{\text{freq}} = 0.87$; $+0.50$ pp on Y_3, $P_{\text{freq}} = 0.90$; and -0.44 pp on Y_4, $P_{\text{freq}} = 0.15$), whereas linear two-way fixed effects (B5) is noisy on Y_1 (-0.21 pp) and linear ridge-based double machine learning (B6) attenuates Y_3 ($+0.07$ pp) because it imposes a constant linear outcome model across heterogeneous development tiers.

Table 4: Method benchmark and ablation comparison for a +10 pp university spending increase (Z_1, Scaling A) at the +10-year horizon on the identical common sample ($N = 283$, 112 countries). Posterior means, 95% Bayesian CIs, and posterior probabilities P are shown for the Bayesian tree specifications; frequentist point estimates, 95% confidence intervals, and one-sided p-value complements $P_{\text{freq}} = 1 - p_{\text{one-sided}}$ are shown for B4–B7.

Estimator / Specification (+10y, Z_1)	Total Tax / GDP (Y_1, pp)	Direct Income Tax / GDP (Y_2, pp)	Direct Income Tax Share (Y_3, pp)	Goods Tax Share (Y_4, pp)
Primary: Joint MV-BCF (Full Controls, M2b)	+0.24 ($P = 0.89$) [−0.15, +0.70]	+0.12 ($P = 0.90$) [−0.08, +0.33]	+0.71 ($P = 0.96$) [−0.08, +1.42]	−0.42 ($P = 0.14$) [−1.19, +0.34]
MV-BCF Without Growth Controls (M1b)	+0.30 ($P = 0.95$) [−0.05, +0.65]	+0.16 ($P = 0.96$) [−0.02, +0.34]	+0.54 ($P = 0.92$) [−0.19, +1.33]	−0.65 ($P = 0.04$) [−1.45, +0.10]
B1: Univariate BCFs ($D = 1$)	+0.45 ($P = 0.97$) [−0.004, +0.86]	+0.17 ($P = 0.94$) [−0.03, +0.39]	+0.71 ($P = 0.93$) [−0.21, +1.63]	−0.47 ($P = 0.15$) [−1.38, +0.44]
B2: MV-BCF Without $\hat{z}(X)$	+0.34 ($P = 0.95$) [−0.05, +0.73]	+0.14 ($P = 0.90$) [−0.08, +0.35]	+0.59 ($P = 0.94$) [−0.11, +1.35]	−0.51 ($P = 0.10$) [−1.23, +0.34]
B3: Single-Forest BART	+0.15 ($P = 0.67$) [−0.53, +0.85]	+0.03 ($P = 0.57$) [−0.30, +0.38]	+0.34 ($P = 0.68$) [−0.93, +1.62]	−0.09 ($P = 0.43$) [−1.39, +1.08]
B4: Linear Local Projections	+0.59 ($P_{\text{freq}} = 0.99$) [+0.07, +1.12]	+0.18 ($P_{\text{freq}} = 0.90$) [−0.09, +0.44]	+0.51 ($P_{\text{freq}} = 0.82$) [−0.58, +1.59]	+0.45 ($P_{\text{freq}} = 0.78$) [−0.70, +1.59]
B5: Two-Way Fixed Effects	−0.21 ($P_{\text{freq}} = 0.30$) [−1.01, +0.59]	+0.08 ($P_{\text{freq}} = 0.61$) [−0.50, +0.67]	+0.50 ($P_{\text{freq}} = 0.70$) [−1.40, +2.41]	−0.32 ($P_{\text{freq}} = 0.37$) [−2.27, +1.63]
B6: Cross-Fitted Double ML	+0.41 ($P_{\text{freq}} = 0.95$) [−0.09, +0.91]	+0.06 ($P_{\text{freq}} = 0.69$) [−0.18, +0.29]	+0.07 ($P_{\text{freq}} = 0.56$) [−0.94, +1.08]	+0.56 ($P_{\text{freq}} = 0.80$) [−0.73, +1.84]
B7: Orthogonal Causal Forest (R-Learner)	+0.25 ($P_{\text{freq}} = 0.92$) [−0.11, +0.62]	+0.11 ($P_{\text{freq}} = 0.87$) [−0.08, +0.30]	+0.50 ($P_{\text{freq}} = 0.90$) [−0.26, +1.26]	−0.44 ($P_{\text{freq}} = 0.15$) [−1.26, +0.38]

Table 5 reports 17 sensitivity and falsification checks (R1 to R17) designed to test every data-construction and modelling decision in our pipeline (Note also that under five-year within-country funding changes Z_2 in Section 5.2, where baseline tax levels average $\bar{L}_1 = 18.04\%$ of GDP and $\bar{L}_3 = 34.68\%$ of revenue across $N = 176$, the first-order level differential $\tau_{Y_2} \approx (\bar{L}_1 \tau_{Y_3} + \bar{L}_3 \tau_{Y_1})/100$ explains why a +1.56 pp shift in Y_3 combined with positive within-country covariance between baseline tax capacity $L_{1,i,t}$ and $\tau_{Y_3,i}$ raises Y_2 by +0.37 pp even when $\tau_{Y_1} = -0.29$ pp). First, the +10-year positive slopes on Y_1, Y_2, and Y_3 hold when we use raw unadjusted WDI tax-to-GDP without resource rent deductions (R1, and +0.24 pp on Y_1, $P = 0.89$, and +0.70 pp on Y_3, $P = 0.96$ when the 150% pre-winsorisation cap is omitted in R1b), collapse the panel into non-overlapping 10-year decadal waves (R2, +0.82 pp on Y_3, $P = 0.93$), sweep out time-invariant country fixed effects via within-country demeaning (R3a at +5 years, +0.44 pp on Y_3, $P = 0.97$; and R3b at +10 years, +0.12 pp on Y_2, $P = 0.94$, and +0.62 pp on Y_3, 95% CI [+0.01, +1.31], $P = 0.98$), or drop all observations with any missing baseline covariate (R4, +0.26 pp on Y_1, $P = 0.91$, and +0.66 pp on Y_3, $P = 0.96$). Second, our pre-treatment falsification test (R5), which checks whether university funding Z_1 at wave t "predicts" prior tax changes over $t - 1 \to t$, yields near-zero slopes (+0.04 pp on Y_2, $P = 0.67$, and +0.001 pp on Y_3, $P = 0.51$), reassuring us that treatment is not tracking

pre-existing fiscal booms. Third, positive 10-year effects hold when we use a more diffuse τ tree prior (R6, +0.30 pp on Y_1, $P = 0.96$, and +0.16 pp on Y_2, $P = 0.97$), perturb SWIID Gini inequality with its standard errors (R7), use row-wise out-of-bag random forest predictions for $\hat{z}(X)$ (R8), apply two-stage sequential g-estimation (R9; Acharya et al. 2016, where Stage 1 regresses $Y_{i,t+2}$ on $M_{i,t\to t+2}$, $Z_{1,i,t}$, and $X_{i,t}$ to demediate the outcome $\tilde{Y}_{i,t+2} = Y_{i,t+2} - M_{i,t\to t+2}\hat{\gamma}$ before fitting MV-BCF in Stage 2, yielding +0.46 pp on Y_1, $P = 0.99$, +0.20 pp on Y_2, $P = 0.97$, and +0.85 pp on Y_3, $P = 0.98$), control for lagged pre-treatment tax trends $\Delta Y_{t-1\to t}$ (R10), or replace the goods tax share with goods tax as a percentage of GDP (R11). Fourth, restricting the sample to pre-COVID waves before 2020 (R12), fitting strictly on the 80% training countries (R13, +0.42 pp on Y_1, $P = 0.99$, and +0.22 pp on Y_2, $P = 0.98$), using the legacy binary median cutoff (R14, $Z \in \{0, 1\}$), feeding the identical covariate matrix into both $\mu(\cdot)$ and $\tau(\cdot)$ (R15, $W = X$, +0.33 pp on Y_1, $P = 0.97$, and +0.51 pp on Y_3, $P = 0.91$), centring treatment at zero in $[-1.5, +1.5]$ (R16, +0.32 pp on Y_1, $P = 0.93$, +0.14 pp on Y_2, $P = 0.90$, and +0.50 pp on Y_3, $P = 0.89$), or estimating the aggregate university budget share Z_4 on the exact overlapping sample with Z_1 and Z_3 (R17, $N = 272$, which still yields negative tax-to-GDP slopes of −0.34 pp on Y_1 and −0.15 pp on Y_2, confirming that Z_4's weak tax-to-GDP return is not an artefact of sample composition) all reinforce our conclusions.

Table 5: Robustness, identification, and falsification checks across the primary tax outcomes (Y_1, Y_2, and Y_3, with 95% CIs on the second line of each cell and posterior probabilities $P = \Pr(\tau > 0 \mid \text{data})$). Checks R1–R13, R15, and R16 report continuous slopes per +10 pp funding increase (Z_1); R14 reports the legacy binary median-cutoff contrast ($Z \in \{0, 1\}$); and R17 reports the aggregate university budget share (Z_4) on the common sample.

Sensitivity Check	Horizon (N, Countries)	Total Tax / GDP (Y_1, pp)	Direct Income Tax / GDP (Y_2, pp)	Direct Income Tax Share (Y_3, pp)
R1. Unadjusted WDI Tax / GDP	+10y (283, 112)	+0.25 ($P = 0.89$) [−0.16, +0.63]	+0.15 ($P = 0.92$) [−0.07, +0.38]	+0.45 ($P = 0.84$) [−0.34, +1.28]
R2. Non-Overlapping 10y Waves	+10y (154, 105)	+0.32 ($P = 0.89$) [−0.17, +0.85]	+0.12 ($P = 0.79$) [−0.16, +0.40]	+0.82 ($P = 0.93$) [−0.30, +1.83]
R3a. Country Fixed Effects (+5y)	+5y (371, 121)	+0.01 ($P = 0.53$) [−0.21, +0.23]	+0.06 ($P = 0.86$) [−0.05, +0.17]	+0.44 ($P = 0.97$) [−0.03, +0.90]
R3b. Country Fixed Effects (+10y)	+10y (283, 112)	+0.18 ($P = 0.86$) [−0.17, +0.50]	+0.12 ($P = 0.94$) [−0.04, +0.26]	+0.62 ($P = 0.98$) [+0.01, +1.31]
R4. Strictly Observed Covariates	+10y (238, 107)	+0.26 ($P = 0.91$) [−0.11, +0.63]	+0.11 ($P = 0.84$) [−0.09, +0.35]	+0.66 ($P = 0.96$) [−0.06, +1.47]
R5. Placebo ($\Delta Y_{t-1\to t}$)	Pre-t (347, 117)	+0.09 ($P = 0.74$) [−0.19, +0.36]	+0.04 ($P = 0.67$) [−0.11, +0.18]	+0.00 ($P = 0.51$) [−0.59, +0.56]
R6. Diffuse τ Prior	+10y (283, 112)	+0.30 ($P = 0.96$) [−0.02, +0.67]	+0.16 ($P = 0.97$) [−0.01, +0.34]	+0.61 ($P = 0.94$) [−0.17, +1.42]
R7. SWIID Error Perturbation	+10y (283, 112)	+0.33 ($P = 0.95$) [−0.06, +0.68]	+0.14 ($P = 0.89$) [−0.09, +0.35]	+0.55 ($P = 0.93$) [−0.20, +1.29]
R8. Row-Wise OOB RF $\hat{z}(X)$	+10y (283, 112)	+0.31 ($P = 0.93$) [−0.10, +0.69]	+0.13 ($P = 0.90$) [−0.08, +0.33]	+0.62 ($P = 0.94$) [−0.19, +1.40]
R9. Sequential G-Estimation	+10y (283, 112)	+0.46 ($P = 0.99$) [+0.09, +0.85]	+0.20 ($P = 0.97$) [−0.01, +0.41]	+0.85 ($P = 0.98$) [+0.05, +1.66]
R10. Pre-Trend Controls ($\Delta Y_{t-1\to t}$)	+10y (283, 112)	+0.28 ($P = 0.96$) [−0.03, +0.61]	+0.16 ($P = 0.94$) [−0.03, +0.36]	+0.58 ($P = 0.93$) [−0.18, +1.42]
R11. Goods Tax / GDP Check	+10y (283, 112)	+0.30 ($P = 0.95$) [−0.04, +0.63]	+0.15 ($P = 0.92$) [−0.06, +0.35]	+0.50 ($P = 0.90$) [−0.30, +1.37]
R12. Pre-COVID Sample (< 2020)	+10y (187, 90)	+0.23 ($P = 0.82$) [−0.27, +0.70]	+0.08 ($P = 0.70$) [−0.20, +0.36]	+0.46 ($P = 0.82$) [−0.49, +1.44]
R13. 80% Training Set Only	+10y (229, 90)	+0.42 ($P = 0.99$) [+0.06, +0.79]	+0.22 ($P = 0.98$) [+0.001, +0.44]	+0.64 ($P = 0.92$) [−0.31, +1.63]
R14. Legacy Binary Cutoff ($Z \in \{0, 1\}$)	+10y (283, 112)	+0.58 ($P = 0.93$) [−0.22, +1.37]	+0.27 ($P = 0.92$) [−0.10, +0.68]	+1.18 ($P = 0.95$) [−0.25, +2.62]
R15. Same Covariates in μ and τ ($W = X$)	+10y (283, 112)	+0.33 ($P = 0.97$) [−0.01, +0.68]	+0.12 ($P = 0.85$) [−0.09, +0.33]	+0.51 ($P = 0.91$) [−0.23, +1.23]
R16. Zero-Centred Scaling ($Z_1 - 1.5$)	+10y (283, 112)	+0.32 ($P = 0.93$) [−0.09, +0.71]	+0.14 ($P = 0.90$) [−0.06, +0.33]	+0.50 ($P = 0.89$) [−0.31, +1.26]
R17. Common-Sample Budget Share (Z_4)	+10y (272, 109)	−0.34 ($P = 0.07$) [−0.72, +0.08]	−0.15 ($P = 0.10$) [−0.37, +0.08]	+0.59 ($P = 0.91$) [−0.26, +1.43]

Finally, we evaluate all 44 countries from our 10-year modelling sample that sit below their income-group training median ($Z_{1,i,t} < 0$) in the most recent treatment wave (2015–2019, saved in full in outputs/tables/tableA4_latest_z0_countries_costs_and_returns.csv alongside the cohort-level lookup in Appendix F, Table 10). Six countries (New Zealand, Iceland, Uruguay, Romania, Italy, and

Slovenia) sit within 2.5 pp of their income-group median (26.4% for high income), illustrating why our continuous MV-BCF specification is much more realistic for country-level policy evaluation than a binary cutoff model. Because the continuous model scales expected returns smoothly with the size of the funding increase, closing a tiny gap to the median (such as +0.31 pp in New Zealand costing €0.04 billion per year or +0.35 pp in Iceland costing €0.005 billion per year) yields a proportionally tiny single-year return at Year 10 (€+0.004 billion and €+0.001 billion per year), whereas a full +10 pp expansion costs €1.16 billion in New Zealand and €0.13 billion in Iceland and yields +0.06 to +0.09 pp of GDP in direct income tax at Year 10 (€+0.14 billion and €+0.02 billion per year) alongside a +0.72 to +0.77 pp shift in Y_3 (note that non-OECD high-income city-states Singapore and Macao SAR fall into a distinct terminal leaf in $W_{i,t}$ with log PPP GDP per person above 11.4 and working-age population share above 73%, yielding higher Y_2 slopes of +0.61 and +0.38 pp). For countries sitting roughly 10 pp below their income-group median (such as Ireland at 17.3% with a 9.1 pp gap costing 0.39% of GDP or €2.16 billion per year to close vs €2.37 billion for a +10 pp increase, South Korea at 14.9%, Argentina at 16.7%, and Belarus at 17.0%), closing the gap and adding +10 pp represent comparable cohort investments whose full 40-year career returns are detailed in Appendix F.

D Continuous Treatment Orthogonalisation, Residual Covariance, and MCMC Diagnostics

Table 6 summarises how our random forest predicted funding level $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ orthogonalises the continuous university funding variable against all 19 pre-treatment covariates across the +10-year modelling sample ($N = 283$, with missing auxiliary covariates imputed via training-fold medians). Before conditioning on $\hat{z}(X_{i,t})$, country-waves with above-median university funding ($Z_{1,i,t} \geq 0$) have higher total education spending as a share of GDP (5.26% vs 3.84%, raw correlation $r = +0.540$), higher baseline non-resource tax-to-GDP (18.57% vs 15.65%, $r = +0.325$), higher baseline direct income tax share (36.57% vs 32.93%, $r = +0.196$), and slower lagged GDP per person growth (1.53% vs 3.49%, $r = -0.297$) than country-waves below the median ($Z_{1,i,t} < 0$). After residualising $Z_{i,t}$ on the random forest prediction $\hat{z}(X_{i,t})$, the mean absolute correlation between the residual funding level $Z_{i,t} - \hat{z}(X_{i,t})$ and all 19 starting covariates drops by more than 60% from 0.157 (maximum 0.540) to 0.061 (maximum 0.213 for Scaling A, and from 0.140 to 0.056 for Scaling B; note that because the random forest orthogonalises $Z_{i,t}$ jointly against all 19 correlated predictors rather than one by one, a single variable with a small raw correlation such as gross tertiary enrolment can shift slightly from −0.090 to −0.131 while the strong confounders are sharply reduced). This confirms that the random forest first stage captures the systematic pre-treatment predictors of funding intensity and that including $\hat{z}(X_{i,t})$ inside $\mu(\cdot)$ orthogonalises the treatment variable against baseline confounders.

Table 6: Continuous treatment covariate correlations before and after residualising on the random forest predicted funding level $\hat{z}(X_{i,t}) = \mathbb{E}[Z_{i,t} \mid X_{i,t}]$ across six representative baseline predictors and the 19-covariate summary (+10-year sample, $N = 283$). Columns 2 and 3 report subgroup means below and above the Scaling A median ($Z_1 = 0$), Columns 4 and 5 report raw and residual correlations with Z_1 (Scaling A), and Columns 6 and 7 report raw and residual correlations with $Z^{(B)}$ (Scaling B).

Baseline Pre-Treatment Covariate	Below Med. ($Z_1 < 0$)	Above Med. ($Z_1 \geq 0$)	Raw Corr. (Scaling A)	Resid. Corr. (Scaling A)	Raw Corr. (Scaling B)	Resid. Corr. (Scaling B)
Total Education Spend (% GDP)	3.84	5.26	+0.540	+0.166	+0.516	+0.179
Baseline Non-Resource Tax / GDP (%)	15.65	18.57	+0.325	+0.044	+0.275	+0.037
Lagged GDP per Person Growth (%)	3.49	1.53	−0.297	−0.134	−0.317	−0.138
Baseline Direct Income Tax Share (%)	32.93	36.57	+0.196	+0.028	+0.176	+0.015
Log Real GDP per Person (PPP)	9.75	9.74	+0.142	+0.025	+0.074	−0.015
Gross Tertiary Enrolment (%)	45.33	36.23	−0.090	−0.131	−0.119	−0.127
All 19 Covariates (Mean / Max $\lvert r \rvert$)	–	–	**0.157 / 0.540**	**0.061 / 0.213**	**0.140 / 0.516**	**0.056 / 0.202**

Table 7 reports the posterior mean residual standard deviations and all six cross-outcome residual correlations from the continuous-treatment MV-BCF model. These posterior correlations show why modelling all four tax outcomes (Y_1 to Y_4) jointly in a Multivariate BCF is preferable to fitting four

independent regressions. Unobserved fiscal shocks to Y_1 (total non-resource tax/GDP) and Y_2 (direct income tax/GDP) are strongly positively correlated ($r = +0.62$ at +5 years and +0.65 at +10 years), as are shocks to Y_2 and Y_3 (the direct income tax share, $r = +0.71$ and +0.75). Conversely, shocks to Y_3 and Y_4 (the goods and services tax share) are strongly negatively correlated ($r = -0.46$ at +5 years and -0.59 at +10 years) because the two revenue shares compete within the same national budget. Estimating the full 4×4 covariance matrix Σ via the Inverse-Wishart Gibbs step allows the continuous tree updates to borrow strength across these correlated fiscal innovations.

Table 7: Posterior mean cross-outcome residual correlations (with 95% CIs) and residual standard deviations from the four-outcome continuous-treatment Multivariate Bayesian Causal Forest (Z_1, Scaling A).

Residual Parameter Pair (MV-BCF, Z_1)	+5y Mean [95% CI]	+10y Mean [95% CI]
1. Non-Res. Tax / GDP (Y_1) ↔ Income Tax / GDP (Y_2)	+0.62 [+0.54, +0.69]	+0.65 [+0.56, +0.73]
2. Income Tax / GDP (Y_2) ↔ Income Tax Share (Y_3)	+0.71 [+0.65, +0.77]	+0.75 [+0.67, +0.81]
3. Income Tax Share (Y_3) ↔ Goods Tax Share (Y_4)	−0.46 [−0.56, −0.36]	−0.59 [−0.69, −0.48]
4. Income Tax / GDP (Y_2) ↔ Goods Tax Share (Y_4)	−0.24 [−0.35, −0.12]	−0.25 [−0.40, −0.11]
5. Non-Res. Tax / GDP (Y_1) ↔ Goods Tax Share (Y_4)	+0.20 [+0.07, +0.32]	+0.33 [+0.19, +0.46]
6. Non-Res. Tax / GDP (Y_1) ↔ Income Tax Share (Y_3)	−0.03 [−0.14, +0.09]	+0.08 [−0.07, +0.23]
7. Residual SDs ($\sigma_1, \sigma_2, \sigma_3, \sigma_4$)	1.80, 1.00, 4.08, 4.21 pp	2.51, 1.39, 5.14, 4.80 pp

Table 8 and Figure 8 present the post-burn-in MCMC convergence diagnostics and posterior predictive check (PPC) fit metrics across the +5-year and +10-year continuous-treatment MV-BCF models (1,250 retained draws). Across all eight horizon-by-outcome parameters in Table 8, split-chain $\hat{R}$ values lie between 0.999 and 1.030 (well below the 1.05 convergence threshold), effective sample sizes (ESS) range from 65 to 120, and Monte Carlo standard errors (MCSE) are small (0.006 to 0.046 pp, less than 13% of the corresponding posterior standard deviations; accounting for within-country clustering across repeated waves leaves the posterior probabilities virtually unchanged at $P_{\text{clu}} = 0.896$ on Y_1, 0.896 on Y_2, and 0.961 on Y_3 at +10 years). Furthermore, in-sample posterior predictive check R^2 values range from 0.247 to 0.641 across the macro-panel difference outcomes, empirical 95% posterior predictive interval coverage (`Cov95`) lies between 95.4% and 97.9%, and 93.8% to 95.8% of standardised residuals fall within ±2 standard deviations. Together with the stationary, well-mixed trace plots in Figure 8 (shown for both Scaling A and Scaling B at +5 and +10 years), these diagnostics confirm that our C++ Gibbs sampler converges reliably and fits the observed distribution of fiscal changes without bias.

Table 8: Post-burn-in MCMC convergence diagnostics (split-chain $\hat{R}$, effective sample size ESS, Monte Carlo standard error MCSE, and Geweke z-score) and in-sample posterior predictive check (PPC) R^2, 95% CI coverage, and share of standardised residuals within ±2 SD across the +5-year and +10-year continuous MV-BCF models (Z_1, Scaling A).

Model & Outcome (Z_1)	Horizon (N)	Mean (SD)	$\hat{R}$	ESS	MCSE	Geweke	PPC R^2	Cov95	$\lvert e^* \rvert \leq 2$SD
1. MV-BCF Tax / GDP (Y_1)	+5y (371)	−0.055 (0.113)	**1.026**	91	0.012	−1.67	0.396	96.2%	93.8%
2. MV-BCF Inc. Tax / GDP (Y_2)	+5y (371)	+0.053 (0.061)	**1.030**	92	0.006	+1.80	0.247	96.2%	95.4%
3. MV-BCF Inc. Tax Share (Y_3)	+5y (371)	+0.365 (0.244)	**1.023**	120	0.022	+0.64	0.332	95.4%	94.1%
4. MV-BCF Goods Share (Y_4)	+5y (371)	+0.098 (0.293)	**1.011**	75	0.034	+0.79	0.426	97.0%	94.6%
5. MV-BCF Tax / GDP (Y_1)	+10y (283)	+0.236 (0.200)	**1.003**	65	0.025	−0.29	0.423	95.8%	94.0%
6. MV-BCF Inc. Tax / GDP (Y_2)	+10y (283)	+0.124 (0.103)	**1.002**	109	0.010	−0.34	0.299	96.5%	95.4%
7. MV-BCF Inc. Tax Share (Y_3)	+10y (283)	+0.707 (0.385)	**1.024**	100	0.038	−1.24	0.460	96.8%	94.7%
8. MV-BCF Goods Share (Y_4)	+10y (283)	−0.422 (0.390)	**0.999**	72	0.046	−1.75	0.641	97.9%	95.8%

E Comparison of Linear Ratio Scaling A and Relative Logarithmic Scaling B

As discussed in Section 3.3, the UNESCO per-student university spending ratio can be transformed either on a linear percentage-point scale within each World Bank income group (Z_1 or Scaling A: $\Delta Z_1 = +1.0$

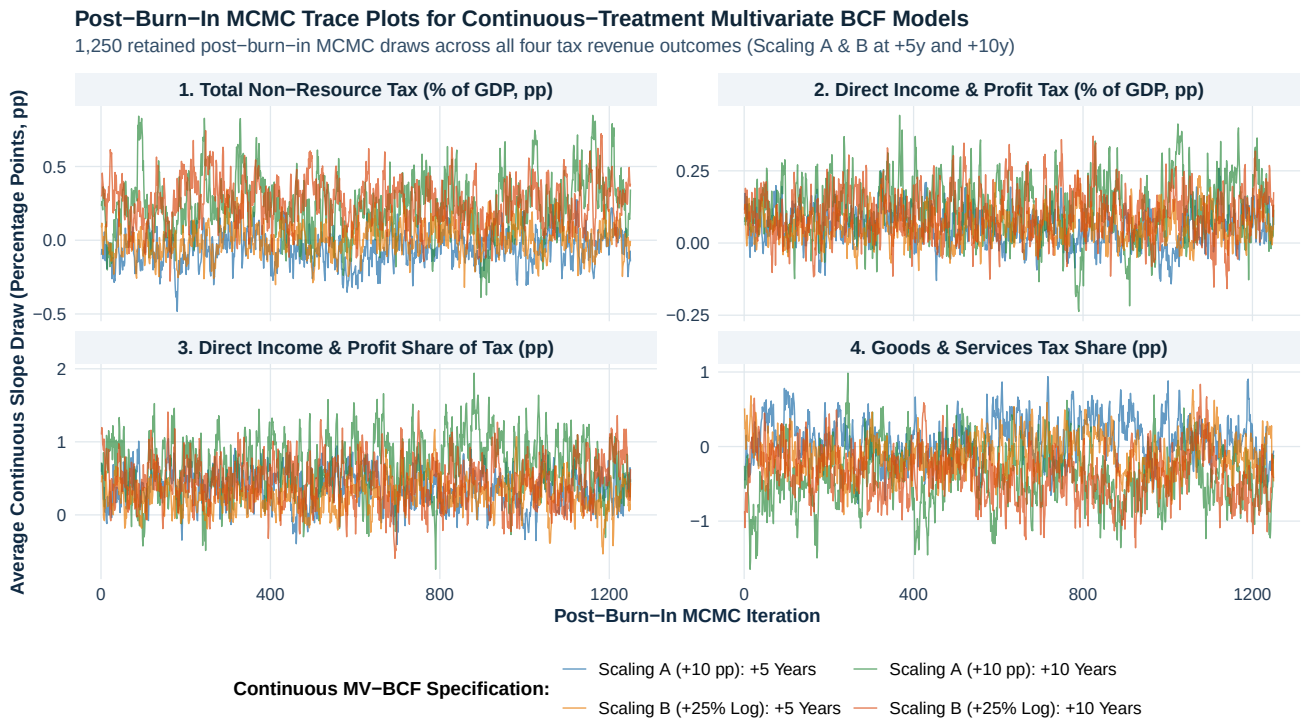


Figure 8: Post-burn-in MCMC trace plots (1,250 retained draws) across all four tax outcomes (Y_1 to Y_4) from the +5y and +10y continuous-treatment Multivariate Bayesian Causal Forest sampler under both Linear Ratio Scaling A (Z_1, +10 pp) and Relative Logarithmic Scaling B (+25% log).

per +10 pp of GDP per person per student around the training-set group median $m_{g(i,t)}$, winsorised at ± 1.5 units so $Z_{1,i,t} \in [-1.5, +1.5]$ with $\mathrm{SD}(Z_1) = 1.05$) or on a concave relative logarithmic scale (Scaling B: $Z_{i,t}^{(\mathrm{B})} = \mathrm{clip}\big(\log(\mathrm{Spend}_{i,t}/m_{g(i,t)})/\log(1.25), -1.5, +1.5\big) \in [-1.5, +1.5]$ with $\mathrm{SD}(Z^{(\mathrm{B})}) = 1.21$, where $\Delta Z^{(\mathrm{B})} = +1.0$ represents a +25% relative increase in per-student funding). In high- and upper-middle-income countries (where median per-student spending is 26.4% and 27.6% of GDP per person), a +25% relative increase equals roughly +6.6 to +6.9 pp of GDP per person. In lower-middle- and low-income countries (where raw medians are 68.9% and 183.8%), a +25% relative increase corresponds to +17.2 to +45.9 pp and compresses the right tail of the UNESCO ratio distribution via the logarithm, directly embedding diminishing marginal returns to additional percentage points of spending. In the main text we present Scaling A (Z_1) because percentage-point increments of GDP per person translate directly into national budget costs; here we compare Scaling A and Scaling B head to head across posterior slopes, in-sample residual diagnostics across $Z \in [-1.5, +1.5]$, 5-fold country-blocked cross-validation, and out-of-sample holdout predictions on the 22 unseen countries in Fold 5.

Table 9: Head-to-head comparison of Linear Ratio Scaling A (Z_1, per +10 pp of GDP per person per student) and Relative Logarithmic Scaling B (per +25% log per-student funding increase) across +5y and +10y posterior slopes (with 95% CIs), in-sample posterior predictive fit (R^2, CRPS, and 95% coverage), 5-fold country-blocked CV across all 112 countries, and locked holdout validation on the 22 unseen countries in Fold 5.

Comparison Metric & Horizon	1. Total Non-Res. Tax / GDP (Y_1, pp)	2. Direct Income Tax / GDP (Y_2, pp)	3. Direct Income Tax Share (Y_3, pp)	4. Goods & Serv. Tax Share (Y_4, pp)
A1. Scaling A Direct (+5y, `M2a`)	-0.06 ($P = 0.31$) $[-0.28, +0.18]$	$+0.05$ ($P = 0.81$) $[-0.07, +0.17]$	$+0.36$ ($P = 0.93$) $[-0.13, +0.81]$	$+0.10$ ($P = 0.62$) $[-0.47, +0.67]$
B1. Scaling B Direct (+5y, `M2c`)	$+0.004$ ($P = 0.50$) $[-0.22, +0.22]$	$+0.06$ ($P = 0.86$) $[-0.05, +0.17]$	$+0.27$ ($P = 0.89$) $[-0.15, +0.73]$	-0.10 ($P = 0.34$) $[-0.58, +0.39]$
A2. Scaling A Direct (+10y, `M2b`)	$+0.24$ ($P = 0.89$) $[-0.15, +0.70]$	$+0.12$ ($P = 0.90$) $[-0.08, +0.33]$	$+0.71$ ($P = 0.96$) $[-0.08, +1.42]$	-0.42 ($P = 0.14$) $[-1.19, +0.34]$
B2. Scaling B Direct (+10y, `M2d`)	$+0.26$ ($P = 0.95$) $[-0.04, +0.56]$	$+0.11$ ($P = 0.90$) $[-0.06, +0.27]$	$+0.47$ ($P = 0.93$) $[-0.16, +1.09]$	-0.38 ($P = 0.13$) $[-0.99, +0.33]$
A3. Scaling A Total (+10y, `M1b`)	$+0.30$ ($P = 0.95$)	$+0.16$ ($P = 0.96$)	$+0.54$ ($P = 0.92$)	-0.65 ($P = 0.04$)
B3. Scaling B Total (+10y, `M1d`)	$+0.21$ ($P = 0.93$)	$+0.11$ ($P = 0.90$)	$+0.41$ ($P = 0.89$)	-0.42 ($P = 0.08$)
A4. Scaling A In-Sample (+10y)	$R^2 = 0.42$, CRPS $= 1.27$ (95.8%)	$R^2 = 0.30$, CRPS $= 0.72$ (96.5%)	$R^2 = 0.46$, CRPS $= 2.67$ (96.8%)	$R^2 = 0.64$, CRPS $= 2.38$ (97.9%)
B4. Scaling B In-Sample (+10y)	$R^2 = 0.49$, CRPS $= 1.20$ (95.8%)	$R^2 = 0.29$, CRPS $= 0.72$ (95.8%)	$R^2 = 0.47$, CRPS $= 2.64$ (97.2%)	$R^2 = 0.63$, CRPS $= 2.42$ (97.9%)
A5. Scaling A 5-Fold CV ($N = 283$)	CRPS $= 1.64$, RMSE $= 3.15$	CRPS $= 0.85$, RMSE $= 1.54$	CRPS $= 3.41$, RMSE $= 6.11$	CRPS $= 3.43$, RMSE $= 6.49$
B5. Scaling B 5-Fold CV ($N = 283$)	CRPS $= 1.64$, RMSE $= 3.16$	CRPS $= 0.86$, RMSE $= 1.56$	CRPS $= 3.51$, RMSE $= 6.32$	CRPS $= 3.47$, RMSE $= 6.59$
A6. Scaling A Holdout ($N = 54$)	CRPS $= 1.73$ ($\bar{\tau}_{\text{test}} = +0.44$)	CRPS $= 0.86$ ($\bar{\tau}_{\text{test}} = +0.21$)	CRPS $= 3.24$ ($\bar{\tau}_{\text{test}} = +0.72$)	CRPS $= 3.08$ ($\bar{\tau}_{\text{test}} = -0.36$)
B6. Scaling B Holdout ($N = 54$)	CRPS $= 1.68$ ($\bar{\tau}_{\text{test}} = +0.31$)	CRPS $= 0.83$ ($\bar{\tau}_{\text{test}} = +0.19$)	CRPS $= 3.15$ ($\bar{\tau}_{\text{test}} = +0.48$)	CRPS $= 3.26$ ($\bar{\tau}_{\text{test}} = -0.49$)

Table 9 and Figure 9 highlight three insights from comparing the two scalings. First, both scalings tell

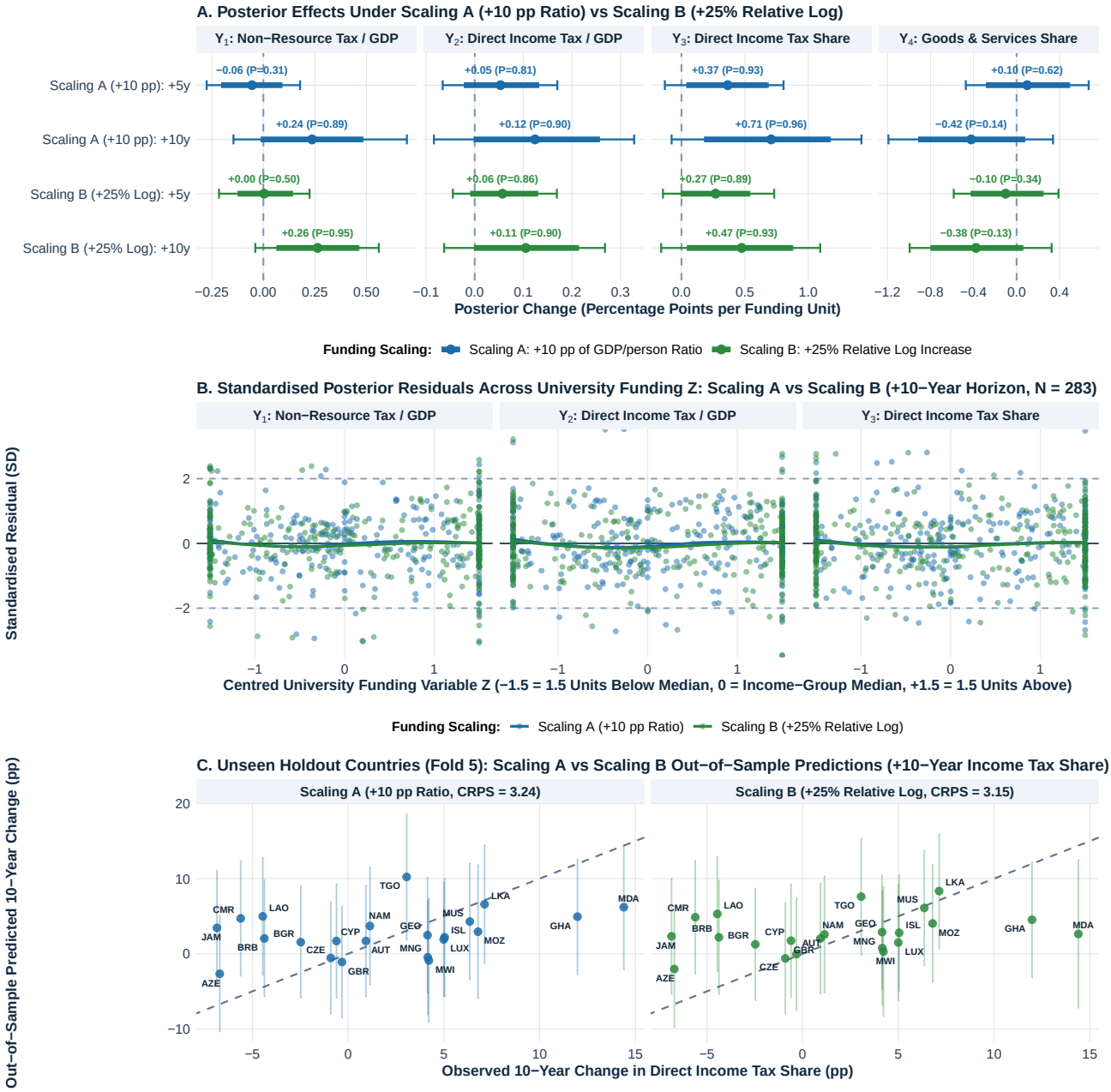


Figure 9: Head-to-head comparison of Linear Ratio Scaling A (Z_1, per +10 pp of GDP per person per student) vs Relative Logarithmic Scaling B (per +25% relative per-student increase): (A) +5y and +10y posterior slopes with 80% and 95% CIs, (B) standardised +10y posterior residuals across $Z \in [-1.5, +1.5]$ with LOESS smooths for both scalings, and (C) observed (horizontal axis) vs out-of-sample predicted (vertical axis, with vertical 80% predictive interval bars) 10-year changes in the direct income tax share (Y_3) across the 22 unseen holdout countries in Fold 5.

the exact same substantive story across horizons and outcomes (Rows A1–B3 and Figure 9A): under both Scaling A and Scaling B, the 5-year effect appears in the direct income tax share ($Y_3 = +0.36$ pp, $P = 0.93$ under Scaling A; +0.27 pp, $P = 0.89$ under Scaling B, and +0.40 pp, $P = 0.97$ for the total effect `M1c`), whilst the 10-year horizon shows gains across Y_1 (+0.24 pp, $P = 0.89$ under Scaling A; +0.26 pp, $P = 0.95$ under Scaling B), Y_2 (+0.12 pp, $P = 0.90$ under Scaling A; +0.11 pp, $P = 0.90$ under Scaling B), and Y_3 (+0.71 pp, $P = 0.96$ under Scaling A; +0.47 pp, $P = 0.93$ under Scaling B). Second, in Figure 9B, the LOESS residual curves across $Z \in [-1.5, +1.5]$ are flat and centred on zero for both scalings, and because a +10 pp increment represents a larger proportional step in high- and upper-middle-income countries, Scaling A achieves slightly lower 5-fold cross-validation error across all 112 countries (Rows A5 vs B5: mean CRPS 2.33 vs 2.37 pp). Third, because the logarithm in Scaling B compresses the right tail of the UNESCO ratio in low- and lower-middle-income countries, Scaling B achieves higher +10-year posterior precision on Y_1 (SD = 0.154 vs 0.200 pp, raising P from 0.89 to 0.95) and slightly lower holdout prediction error on the 22 unseen countries in Fold 5 for Y_1, Y_2, and Y_3 (Row B6 and Figure 9C: CRPS 1.68 vs 1.73 pp on Y_1, 0.83 vs 0.86 pp on Y_2, and 3.15 vs 3.24 pp on Y_3, with Diebold-Mariano $p = 0.82$–0.84).

F Assistance for Policy Makers

To supplement the cohort return-on-investment analysis in Section 5.4, Figure 10 plots each country's current per-student university spending level on the horizontal axis against its estimated 40-year career return-on-investment ratio (ROI = 40-Year Career Tax Output/Cohort Degree Input, with vertical bars indicating 95% credible intervals) for both Total Non-Resource Tax (Y_1, Panel A) and Direct Income Tax (Y_2, Panel B), while Table 10 provides a compact country-by-country lookup guide for 20 representative high-, middle-, and low-income economies from the 70 countries with complete `2020–2024` macroeconomic, demographic, enrolment, and fiscal data (all 70 countries are saved in `outputs/tables/table_policy_maker_assistance_2024.csv` and can be filtered and explored interactively across Z_1 to Z_4 and Y_1 to Y_4 in our companion web application in `app/`). Because UNESCO's disaggregated per-student tertiary series ends in `2018–2019` in the World Bank archive whereas national GDP, population, university enrolment, total education budgets, and tax revenues run through `2024`, we pair each country's `2020–2024` GDP, population, enrolment, and tax base with its latest UNESCO per-student funding ratio to express all cohort inputs and career tax outputs in constant 2021 euros (€ billions; policy makers in countries whose per-student funding or student headcount shifted substantially after `2019` can input updated national baseline figures directly in Tab 3 of the companion web application). As derived in Section 5.4, a single entering cohort makes up $1/D$ of the total student body in any calendar year and studies for D years, so its multi-year degree input cost equals one year of the system-wide +10 pp budget increase, and its cumulative 40-year career tax output equals eight times the Year-10 annual flow (40 working years ÷ 5 cohorts = 8, or 5.7× when discounted at $r - g = 1.5\%$ per year and 4.0× under a conservative two-wave level attribution; multiply both input and career outputs by 5 for a five-year wave of five cohorts). Notice also in Figure 10 and Table 10 that low- and lower-middle-income countries with low tertiary enrolment (such as Bangladesh at 11.66×) exhibit high raw benefit-cost ratios because a +10 pp per-student increase costs a very small share of GDP when enrolment is small (0.19% of GDP in Bangladesh) while $\tau_{10}(W_i)$ is estimated in percentage points of total GDP; scaling those developing-country returns by their enrolment rate relative to the high-income benchmark places their cohort benefit-cost ratios in the same 2.0× to 4.5× range as high-income OECD economies. For treasury budget decisions, policy makers should focus primarily on Total Non-Resource Tax (Y_1), which captures the full Exchequer payback across income, corporate, employer payroll/social insurance, value-added, and property taxes, while using Direct Income Tax (Y_2) and the Direct Income Tax Share shift (Y_3) as a conservative floor that verifies the return comes from higher formal salaries and company profits.

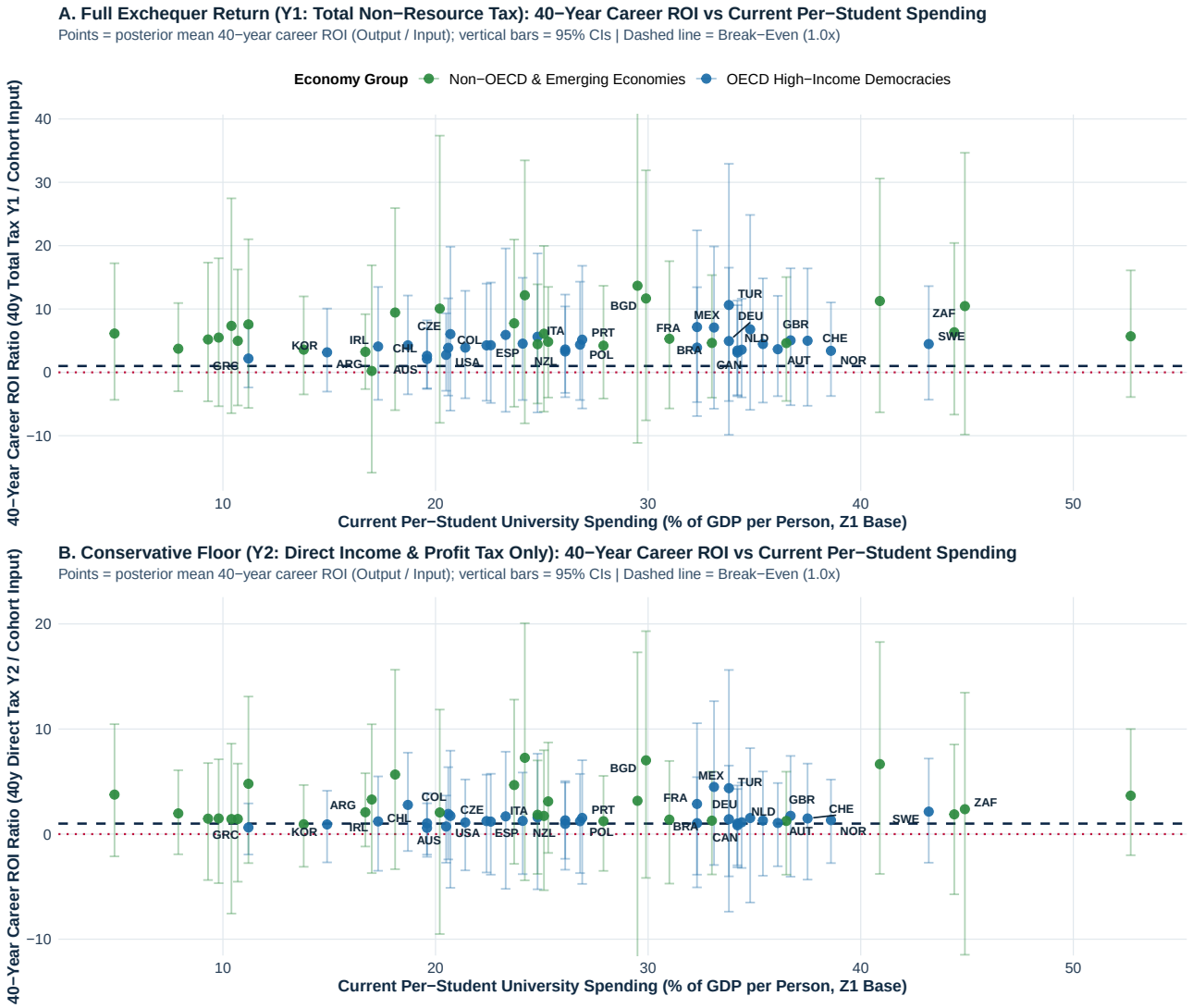


Figure 10: Current per-student university spending (% of GDP per person, Z_1 base, horizontal axis) vs estimated 40-year career Return on Investment ratio (ROI = 40-Year Career Tax Output/Cohort Degree Input, vertical axis, with points showing posterior means and vertical bars showing 95% credible intervals) for (A) Total Non-Resource Tax (Y_1, full Exchequer return) and (B) Direct Income and Profit Tax (Y_2, conservative floor). The horizontal dashed line marks break-even (ROI = 1.0×, where 40-year career tax output equals the cohort degree input cost), and the dotted line marks zero return (0.0×). To keep the axes legible, four sub-Saharan African countries with UNESCO ratios above 60% due to < 5% tertiary enrolment (`BFA`, `CIV`, `RWA`, `TGO`) and four city-state or high-variance cases (`SGP`, `MAC`, `LUX`, `MDA`, `RUS`) are omitted from the plot but included in `outputs/tables/table_policy_maker_assistance_2024.csv` and the interactive web application.

Table 10: Assistance for policy makers across 20 representative countries with `2020–2024` macroeconomic, enrolment, and fiscal data (full 70-country table in `outputs/tables/table_policy_maker_assistance_2024.csv`). Columns report current per-student university spending (% of GDP per person and 2021 PPP dollars), the multi-year degree input cost for a single entering cohort under a +10 pp per-student funding increase ($Z_1 = +1.0$ unit, in % of GDP and € billions; multiply input and career outputs by 5 for a five-year wave), the single-year annual tax flows at +5y and +10y (€B/yr), and the cumulative 40-year career tax output per cohort (€B, with 95% CIs in brackets, benefit-cost ratio BCR = Output/Input, and posterior probability of a net positive return $P_{\text{net}} = \Pr(\text{Output} > \text{Input} \mid \text{data})$) for both Total Non-Resource Tax (Y_1) and Direct Income Tax (Y_2), plus the +10-year shift in the Direct Income Tax Share (Y_3, pp).

Country (Group)	Spend % GDP/pc ($)	Cohort Input	+5y → +10y Y_1/yr	40y Career Output Y_1 [95% CI] (BCR, P_{net})	40y Career Output Y_2 [95% CI] (BCR, P_{net})	+10y Shift Y_3 [95% CI]
Australia (High)	19.6% ($11,199)	0.81% (€11.54B)	€−1.13 → +3.03B	**€**+24.25**B** [€ − 29.1, +80.8B] (2.10×, 0.68)	**€**+6.81**B** [€ − 25.0, +33.8B] (0.59×, 0.38)	+0.72 [−0.19, +1.51] (0.95)
Austria (High)	36.1% ($22,638)	0.48% (€2.56B)	€−0.40 → +1.17B	**€**+9.36**B** [€ − 9.6, +30.9B] (3.65×, 0.76)	**€**+2.72**B** [€ − 7.8, +12.4B] (1.06×, 0.51)	+0.73 [−0.13, +1.51] (0.96)
Canada (High)	34.2% ($19,336)	0.55% (€11.31B)	€−1.60 → +4.42B	**€**+35.36**B** [€ − 42.5, +119.9B] (3.13×, 0.73)	**€**+9.46**B** [€ − 35.7, +48.5B] (0.84×, 0.47)	+0.71 [−0.20, +1.50] (0.94)
Chile (High)	19.6% ($5,491)	0.70% (€3.63B)	€−0.22 → +1.18B	**€**+9.46**B** [€ − 9.5, +29.8B] (2.61×, 0.73)	**€**+3.75**B** [€ − 7.1, +14.2B] (1.03×, 0.51)	+0.76 [−0.10, +1.59] (0.96)
France (High)	32.3% ($16,767)	0.26% (€8.59B)	€+2.98 → +7.67B	**€**+61.37**B** [€ − 59.4, +192.4B] (7.15×, 0.81)	**€**+24.68**B** [€ − 43.6, +90.6B] (2.87×, 0.68)	+0.77 [−0.09, +1.59] (0.96)
Germany (High)	33.8% ($20,773)	0.37% (€17.81B)	€+0.43 → +11.00B	**€**+88.00**B** [€ − 80.8, +294.2B] (4.94×, 0.79)	**€**+25.37**B** [€ − 71.5, +116.1B] (1.42×, 0.56)	+0.74 [−0.12, +1.52] (0.96)
Ireland (High)	17.3% ($15,150)	0.43% (€2.37B)	€−0.24 → +1.21B	**€**+9.66**B** [€ − 10.3, +31.9B] (4.08×, 0.77)	**€**+2.87**B** [€ − 8.3, +13.0B] (1.21×, 0.54)	+0.73 [−0.12, +1.51] (0.96)
Italy (High)	24.8% ($12,061)	0.31% (€8.47B)	€−0.33 → +5.95B	**€**+47.62**B** [€ − 53.6, +159.0B] (5.62×, 0.78)	**€**+13.46**B** [€ − 44.5, +64.8B] (1.59×, 0.57)	+0.72 [−0.16, +1.51] (0.95)
Korea, Rep. (High)	14.9% ($7,062)	0.58% (€14.52B)	€−2.09 → +5.73B	**€**+45.88**B** [€ − 44.1, +146.2B] (3.16×, 0.74)	**€**+13.35**B** [€ − 39.1, +60.0B] (0.92×, 0.48)	+0.74 [−0.14, +1.59] (0.94)
Netherlands (High)	34.4% ($22,547)	0.47% (€5.29B)	€+0.05 → +2.36B	**€**+18.89**B** [€ − 20.9, +61.5B] (3.57×, 0.75)	**€**+5.91**B** [€ − 17.0, +25.9B] (1.12×, 0.52)	+0.74 [−0.12, +1.51] (0.96)
Norway (High)	38.6% ($34,419)	0.53% (€2.48B)	€+0.12 → +1.06B	**€**+8.45**B** [€ − 9.3, +27.4B] (3.40×, 0.75)	**€**+3.29**B** [€ − 6.8, +12.9B] (1.33×, 0.56)	+0.76 [−0.13, +1.59] (0.96)
Poland (High)	26.8% ($9,283)	0.42% (€5.97B)	€−1.06 → +3.26B	**€**+26.10**B** [€ − 26.1, +85.4B] (4.37×, 0.77)	**€**+7.31**B** [€ − 22.2, +34.2B] (1.23×, 0.54)	+0.73 [−0.15, +1.54] (0.95)
Spain (High)	22.4% ($10,110)	0.41% (€8.34B)	€−1.56 → +4.44B	**€**+35.52**B** [€ − 37.4, +116.9B] (4.26×, 0.77)	**€**+10.36**B** [€ − 30.5, +47.2B] (1.24×, 0.54)	+0.73 [−0.13, +1.51] (0.95)
Sweden (High)	43.2% ($25,856)	0.43% (€2.58B)	€+0.19 → +1.44B	**€**+11.53**B** [€ − 11.1, +35.1B] (4.47×, 0.79)	**€**+5.52**B** [€ − 7.0, +18.5B] (2.14×, 0.67)	+0.78 [−0.14, +1.67] (0.95)
United Kingdom (High)	36.7% ($19,099)	0.35% (€11.55B)	€+1.82 → +7.29B	**€**+58.33**B** [€ − 60.0, +189.6B] (5.05×, 0.79)	**€**+20.14**B** [€ − 46.8, +85.9B] (1.74×, 0.60)	+0.75 [−0.11, +1.58] (0.96)
United States (High)	20.5% ($13,612)	0.67% (€148.45B)	€−16.29 → +51.18B	**€**+409.46**B** [€ − 432.1, +1383.6B] (2.76×, 0.72)	**€**+106.27**B** [€ − 404.5, +545.1B] (0.72×, 0.43)	+0.71 [−0.20, +1.52] (0.94)
Brazil (Upper Mid.)	33.0% ($5,924)	0.40% (€14.42B)	€−2.83 → +8.37B	**€**+66.98**B** [€ − 57.7, +221.4B] (4.64×, 0.78)	**€**+18.45**B** [€ − 55.4, +86.0B] (1.28×, 0.54)	+0.73 [−0.15, +1.51] (0.95)
Colombia (Upper Mid.)	20.6% ($3,518)	0.49% (€4.13B)	€+0.21 → +2.00B	**€**+16.00**B** [€ − 15.2, +48.3B] (3.87×, 0.78)	**€**+7.93**B** [€ − 10.0, +26.3B] (1.92×, 0.66)	+0.80 [−0.06, +1.67] (0.97)
Mexico (Upper Mid.)	33.1% ($7,171)	0.29% (€7.39B)	€+1.39 → +6.54B	**€**+52.31**B** [€ − 42.4, +146.8B] (7.08×, 0.84)	**€**+33.16**B** [€ − 21.8, +93.5B] (4.49×, 0.81)	+0.85 [−0.01, +1.81] (0.97)
Bangladesh (Lower Mid.)	29.9% ($1,810)	0.19% (€2.31B)	€+1.84 → +3.36B	**€**+26.89**B** [€ − 17.5, +73.5B] (11.66×, 0.87)	**€**+16.18**B** [€ − 9.6, +44.5B] (7.02×, 0.84)	+0.83 [−0.04, +1.75] (0.97)

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All data processing, C++ (`RcppArmadillo`) Gibbs sampling, country-blocked cross-validation, and figure generation were executed on a single Apple Silicon workstation (`macOS`, R version 4.6.1 [2026-06-24], Apple Clang C++17 compiler). Table 11 reports both the local computational footprint of our statistical modelling pipeline (assuming an average 25 W CPU package power draw during single-core C++ tree sampling) and the cloud inference footprint of the large language model (LLM) queries used to assist with building the data pipeline, writing the C++ sampler and holdout validation suite, conducting multi-agent and external referee reviews, developing the companion web application, and editing the manuscript (evaluated at the Irish grid and European cloud average carbon intensity of 250 g CO_2e/kWh). Because our reversible-jump tree search and conjugate 4×4 Inverse-Wishart covariance updates are written directly in C++, fitting all 68 main, Scaling A vs Scaling B, 5-fold country-blocked CV, holdout, benchmark, and sensitivity models (including Checks `B7`, `R15`, `R16`, and `R17`) completes in roughly 20.8 minutes (1,248.0 seconds) and consumes only 8.67 Wh of electricity (2.17 g CO_2e). Auditing our complete interaction logs across all five project stages reveals 840 LLM queries (comprising 1.85 million input and context tokens and 1.09 million generated output, reasoning, and tool-call tokens across the primary session, eight specialist reviewer and user-persona subagents, and external LLM referee rounds; token counts in Table 11 are rounded to the nearest thousand or hundredth of a million while energy and CO_2e are computed from unrounded totals). Applying published frontier LLM inference energy benchmarks (Luccioni et al. 2024; 0.30 Wh base overhead per agentic query plus 0.40 Wh per 1,000 generated output tokens and 0.05 Wh per 1,000 prefill input tokens), the cloud LLM queries consumed an estimated 780.50 Wh (195.13 g CO_2e), bringing the combined project footprint to 789.17 Wh (197.30 g CO_2e, equivalent to driving a typical petrol passenger car approximately one kilometre).

Table 11: Computational wall-clock time, token volume (rounded to the nearest thousand or hundredth of a million), electrical energy use, and estimated carbon footprint across both the local C++ continuous MV-BCF and cross-validation pipeline and the cloud LLM queries used to help build, review, and write the paper.

Project Stage	Models Fitted / LLM Query & Token Volume	Time (s)	Energy (Wh)	CO_2e (g)
A. Local C++ MCMC & CV Pipeline	68 four-outcome MV-BCF, CV & benchmark fits (2,000 iter)	1,248.0	8.67	2.17
B1. LLM: Scoping, Continuous-Z & Data	58 queries (102k input / 68k output tokens)	–	49.80	12.45
B2. LLM: C++ Sampler, Scaling & Holdout CV	39 queries (64k input / 45k output tokens)	–	33.40	8.35
B3. LLM: Manuscript & Revisions	340 queries (638k input / 394k output tokens)	–	291.90	72.98
B4. LLM: Multi-Agent & Peer Review	261 queries (682k input / 361k output tokens)	–	256.80	64.20
B5. LLM: Policy Web App & Testing	142 queries (362k input / 219k output tokens)	–	148.60	37.15
Subtotal: Cloud LLM Assistance	**840 queries (1.85M input / 1.09M output tokens)**	–	**780.50**	**195.13**
Total Project Footprint (A + B)	**68 C++ MCMC/CV fits + 840 LLM queries (2.94M tokens)**	**1,248.0**	**789.17**	**197.30**

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